

The inscribed polygon

A polygon whose every vertex lies on one circle — and the condition that decides when that is possible.

LESSON 33–34

2 × 40 MIN

GEOMETRIYA 9, §21

IGX 3.5

LESSON CLOCK

13'

23'

22'

18'

4'

The definition	13 min
Every triangle has one	23 min
Cyclic quadrilaterals	22 min
Deciding	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define an inscribed polygon and a circumscribed circle.
- Prove that every triangle has a circumscribed circle, and construct its centre.
- State and use the cyclic-quadrilateral condition $\angle A + \angle C = 180^\circ$.
- Decide whether a given quadrilateral is cyclic.

TEXTBOOK

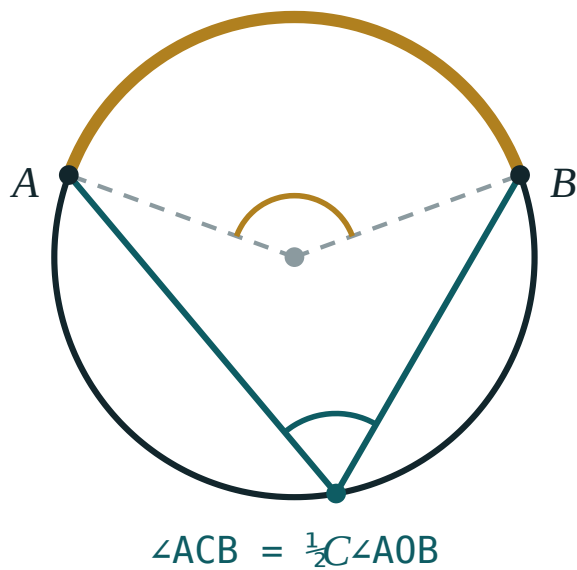
National	pp. 121–127
Cambridge	Core & Extended, pp. 63–70
Workbook	Exercise 3.5

01

Explanation

The definition

A polygon is **inscribed** in a circle if every one of its vertices lies on the circle; the circle is then **circumscribed** about the polygon.



A polygon inscribed in a circle — every vertex on the circumference, every side a chord.

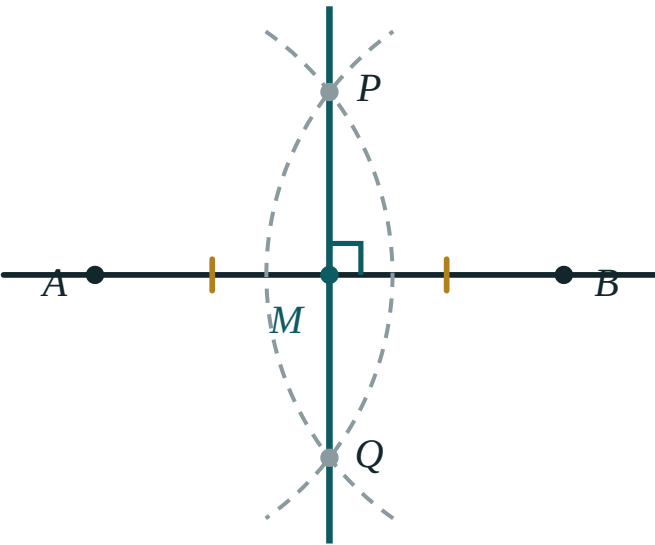
POLYGON	ALWAYS HAS A CIRCUMSCRIBED CIRCLE?
any triangle	yes
a rectangle	yes
a general quadrilateral	no
a parallelogram (not a rectangle)	no
any regular polygon	yes

EVERY SIDE OF AN INSCRIBED POLYGON IS A CHORD

So all the chord theorems of Grade 8 apply at once: equal chords subtend equal angles, the perpendicular from the centre bisects the chord, and the inscribed angle is half the central angle.

Every triangle has one

The three perpendicular bisectors of the sides of a triangle meet in one point, which is equidistant from all three vertices



A point on the perpendicular bisector of a segment is equidistant from its ends.

The proof is short. A point on the perpendicular bisector of AB is equidistant from A and B ; one on the bisector of BC is equidistant from B and C . Their intersection O therefore satisfies $OA = OB = OC$, so it also lies on the third bisector, and the circle of radius OA passes through all three vertices.

TRIANGLE	WHERE THE CIRCUMCENTRE LIES
acute	inside
right	at the midpoint of the hypotenuse
obtuse	outside

The middle row is Thales' theorem restated: the hypotenuse of a right triangle is a diameter of its circumcircle, and $R = \frac{c}{2}$.

Cyclic quadrilaterals

A quadrilateral can be inscribed in a circle if and only if its opposite angles are supplementary

$\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$

The forward direction follows from the inscribed-angle theorem: $\angle A$ and $\angle C$ stand on the two arcs BCD and BAD , whose central angles add to 360° , so the inscribed angles add to 180° .

QUADRILATERAL	CYCLIC?	WHY
a rectangle	yes	all angles 90°
a square	yes	a rectangle
an isosceles trapezium	yes	base angles equal
a rhombus (not a square)	no	opposite angles are equal, not supplementary
a kite	only if two opposite angles are 90°	—

EQUAL IS NOT SUPPLEMENTARY

A parallelogram has $\angle A = \angle C$. That gives $\angle A + \angle C = 180^\circ$ only when both are 90° — which is why the only cyclic parallelograms are rectangles.

Deciding

Three practical tests for whether four points lie on a circle:

TEST	CONDITION
opposite angles	$\angle A + \angle C = 180^\circ$
an exterior angle	equals the opposite interior angle
equal angles on one side	$\angle ACB = \angle ADB$

The third is the converse of “angles on the same arc are equal”, and it is the one used when the quadrilateral is not drawn but two triangles share a base.

CYCLIC QUADRILATERALS APPEAR CONSTANTLY IN CIRCLE PROOFS

Recognising one in a complicated figure converts an unknown angle into a known one in a single step. It is the most useful theorem of the Grade 9 circle work.

02 Worked examples

EXAMPLE 1 A cyclic quadrilateral has $\angle A = 85^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.

① $\angle A + \angle C = 180^\circ$

② $\angle C = 95^\circ$

③ $\angle B + \angle D = 180^\circ$

④ $\angle D = 70^\circ$

Check: $85 + 110 + 95 + 70 = 360$ ✓

ANSWER

$\angle C = 95^\circ, \angle D = 70^\circ$

EXAMPLE 2 A right triangle has legs 6 and 8. Find the radius of its circumscribed circle.

① The hypotenuse is 10.

② For a right triangle the hypotenuse is a diameter.
Thales.

③ $R = \frac{10}{2}$

④ $R = 5$

ANSWER

$R = 5$

EXAMPLE 3 Is a rhombus with angles 60° and 120° cyclic?

① Opposite angles of a rhombus are equal.

② So they are $60^\circ, 60^\circ$ and $120^\circ, 120^\circ$.

③ $60 + 60 = 120 \neq 180$

④ No — it is not cyclic.
Only a square among rhombi is.

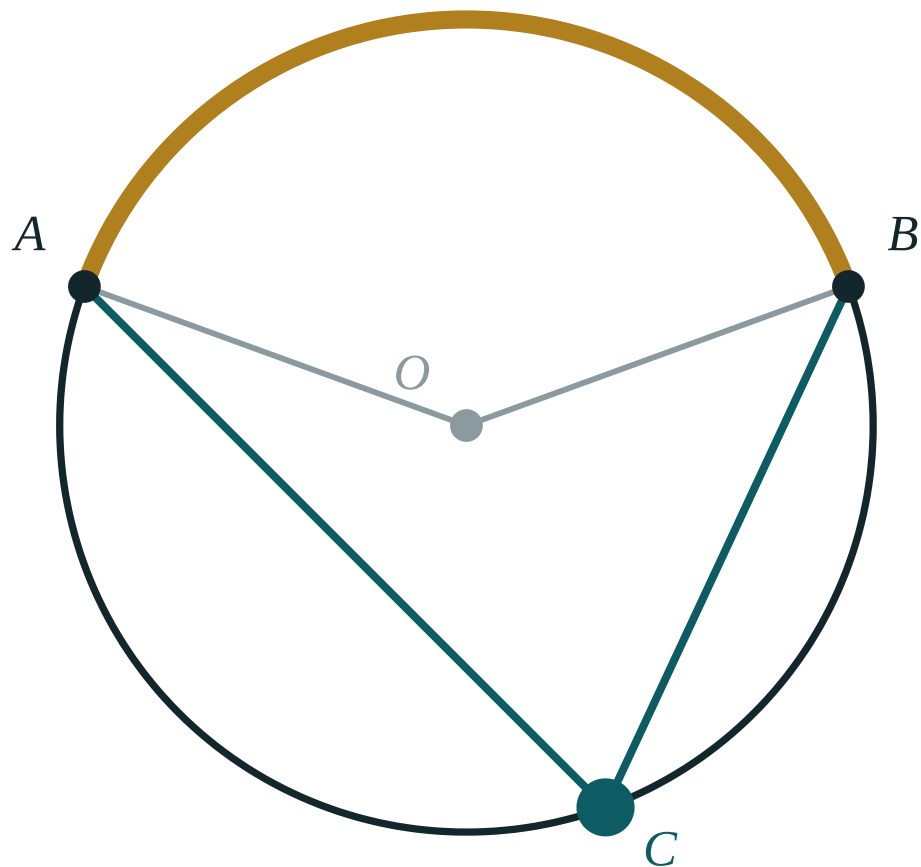
ANSWER

No

03 Interactive model

Draw a triangle on the board and construct two perpendicular bisectors; the third passes through their meeting point without being drawn, and the class sees why.

Four points on a circle



central $\angle AOB$ 140°

inscribed $\angle ACB$ 70°

ratio 2

arc AB 140°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Inscribed polygon	Ichki chizilgan ko'pburchak	Вписанный многоугольник
Circumscribed circle	Tashqi chizilgan aylana	Описанная окружность
Circumcentre	Tashqi markaz	Центр описанной окружности
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Cyclic quadrilateral	Ichki chizilgan to'rtburchak	Вписанный четырёхугольник
Opposite angles	Qarama-qarshi burchaklar	Противоположные углы
Chord	Vatar	Хорда
Concurrent	Bir nuqtada kesishuvchi	Пересекающиеся в одной точке

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An inscribed polygon has:

its sides touching the circle

its vertices on the circle

the circle inside

no circle

The circumcentre is where the _____ meet:

medians

bisectors of the angles

perpendicular bisectors

altitudes

For a right triangle, R equals:

$$\frac{a}{2}$$

$$\frac{c}{2}$$

c

$$\frac{a + b}{2}$$

A quadrilateral is cyclic when:

its sides are equal

opposite angles sum to 180°

its diagonals are equal

always

Which parallelogram is cyclic?

all of them

the rhombus

the rectangle

none

An isosceles trapezium is:

never cyclic

always cyclic

sometimes cyclic

not a quadrilateral

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Cyclic quadrilateral, $\angle A = 85^\circ$: $\angle C$

ANSWER 95°

2. Cyclic quadrilateral, $\angle B = 110^\circ$: $\angle D$

ANSWER 70°

3. Right triangle legs 6, 8: R

ANSWER 5

4. Right triangle legs 5, 12: R

ANSWER 6.5

5. Is a rectangle cyclic?

ANSWER Yes

6. Is a rhombus (not a square) cyclic?

ANSWER No

7. Where is the circumcentre of an acute triangle?

ANSWER Inside

Medium · the standard question

7 PROBLEMS

1. Cyclic, $\angle A = 3x$ and $\angle C = 2x$: x

ANSWER 36°

2. Cyclic, $\angle A = 100^\circ$ and $\angle B = 75^\circ$: the other two

ANSWER 80° and 105°

3. Where is the circumcentre of an obtuse triangle?

ANSWER Outside

4. An equilateral triangle of side 6: R

ANSWER $2\sqrt{3}$

5. Is an isosceles trapezium cyclic?

ANSWER Yes

6. The exterior angle of a cyclic quadrilateral equals

ANSWER The opposite interior angle

7. A triangle 5, 12, 13: R

ANSWER 6.5

Hard · stretch and reason

7 PROBLEMS

1. A cyclic quadrilateral with $\angle A : \angle C = 4 : 5$

ANSWER 80° and 100°

2. A triangle $a = 7, A = 30^\circ$: R

ANSWER 7

3. A cyclic quadrilateral with sides 4, 5, 6, 7: its area (Brahmagupta)

ANSWER ≈ 29.0

4. A square of side 8: the radius of its circumscribed circle

ANSWER $4\sqrt{2}$

5. A regular hexagon of side 6: R

ANSWER 6

6. A cyclic quadrilateral has an exterior angle of 70° : the opposite interior angle

ANSWER 70°

7. Prove that a rectangle is cyclic

ANSWER Opposite angles are both 90°

07 Homework

Homework — 5 tasks

Mark the circle and its centre on every figure before hunting for angles.

1. A cyclic quadrilateral has $\angle A = 72^\circ$ and $\angle B = 95^\circ$. Find the other two angles.
2. A right triangle has legs 9 and 12. Find the radius of its circumscribed circle.
3. Show that a parallelogram is cyclic only if it is a rectangle.
4. Find the circumradius of an equilateral triangle of side 10.
5. A cyclic quadrilateral has $\angle A : \angle C = 2 : 3$. Find both angles.

The circumscribed polygon

A polygon whose every side touches one circle — and the tangent lengths that make it work.

LESSON 35–36

2 × 40 MIN

GEOMETRIYA 9, §22

IGX 3.5

LESSON CLOCK

13'

23'

22'

18'

4'

The definition	13 min
Every triangle has one	23 min
The area formula	22 min
Tangential quadrilaterals	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define a circumscribed polygon and an inscribed circle.
- Prove that every triangle has an inscribed circle and locate its centre.
- Use $S = pr$, where p is the semiperimeter.
- State and use the tangential-quadrilateral condition $a + c = b + d$.

TEXTBOOK

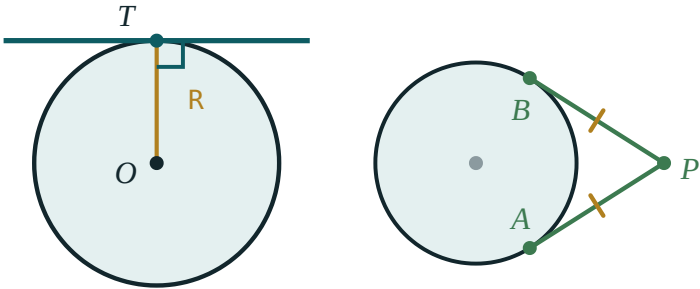
National	pp. 128–134
Cambridge	Core & Extended, pp. 63–70
Workbook	Exercise 3.5

01

Explanation

The definition

A polygon is **circumscribed** about a circle if every one of its sides is tangent to the circle; the circle is then **inscribed** in the polygon.



tangent ⊥ radius at T two tangents from P are equal

A tangent is perpendicular to the radius at the point of contact.

Two facts from Grade 8 are used throughout:

FACT	STATEMENT
tangent and radius	perpendicular at the point of contact
two tangents from one point	equal in length

INSCRIBED AND CIRCUMSCRIBED ARE OPPOSITE WORDS

A polygon inscribed **in** a circle has its vertices on it; a polygon circumscribed **about** a circle has its sides touching it. Reading which is which in a question is worth doing slowly.

Every triangle has one

The three angle bisectors of a triangle meet in one point, which is equidistant from all three sides

A point on the bisector of an angle is equidistant from its two arms. The intersection I of two bisectors is therefore equidistant from all three sides, so the circle centred at I with that distance as radius touches all three sides.

CENTRE	MEETING POINT OF	ALWAYS INSIDE?
circumcentre	the perpendicular bisectors	no
incentre	the angle bisectors	yes
centroid	the medians	yes
orthocentre	the altitudes	no

THE INCENTRE IS ALWAYS INSIDE

Because it is equidistant from the three sides and inside the angle at each vertex. That distinguishes it from the circumcentre, which leaves the triangle as soon as an angle becomes obtuse.

The area formula

Join the incentre to the three vertices. The triangle splits into three triangles, each with height r on one of the sides:

$$S = \frac{1}{2}ar + \frac{1}{2}br + \frac{1}{2}cr = \frac{1}{2}(a + b + c)r$$

$S = pr$, where $p = \frac{a + b + c}{2}$ is the semiperimeter

TRIANGLE	P	S	$R = \frac{S}{P}$
3, 4, 5	6	6	1
5, 12, 13	15	30	2
13, 14, 15	21	84	4

For a right triangle there is a shortcut: $r = \frac{a + b - c}{2}$, which the 3–4–5 row confirms.

P IS THE SEMIPERIMETER

Using the whole perimeter halves the inradius. Writing p out as $\frac{a + b + c}{2}$ once, at the top of the solution, prevents it.

Tangential quadrilaterals

A quadrilateral has an inscribed circle if and only if the sums of its opposite sides are equal

$a + c = b + d$

The proof uses the equal tangent lengths. If the tangent lengths from the four vertices are x, y, z, t , then $a = x + y$, $b = y + z$, $c = z + t$, $d = t + x$, and both sums equal $x + y + z + t$.

QUADRILATERAL	TANGENTIAL?
a rhombus	yes — all sides equal
a square	yes
a kite	yes — $a + c = b + d$ by symmetry
a rectangle (not a square)	no

THE RECTANGLE AND THE RHOMBUS SWAP PLACES

A rectangle is cyclic but not tangential; a rhombus is tangential but not cyclic. Only the square is both — which is the neatest way to remember the two conditions.

02 Worked examples

EXAMPLE 1 Find the inradius of the triangle with sides 13, 14, 15.

$$① \quad p = \frac{13 + 14 + 15}{2} = 21$$

$$② \quad \text{Heron: } S = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84.$$

$$③ \quad r = \frac{S}{p} = \frac{84}{21}$$

$$④ \quad r = 4$$

ANSWER

$$r = 4$$

EXAMPLE 2 A tangential quadrilateral has sides 7, 9, 11 and x in order. Find x .

$$① \quad a + c = b + d$$

$$② \quad 7 + 11 = 9 + x$$

$$③ \quad 18 = 9 + x$$

$$④ \quad x = 9$$

ANSWER

$$x = 9$$

EXAMPLE 3 Find the inradius of a right triangle with legs 6 and 8.

① Hypotenuse 10, $S = 24$, $p = 12$.

② $r = \frac{24}{12} = 2$

③ Check with the shortcut: $\frac{6 + 8 - 10}{2} = 2 \checkmark$

④ $r = 2$

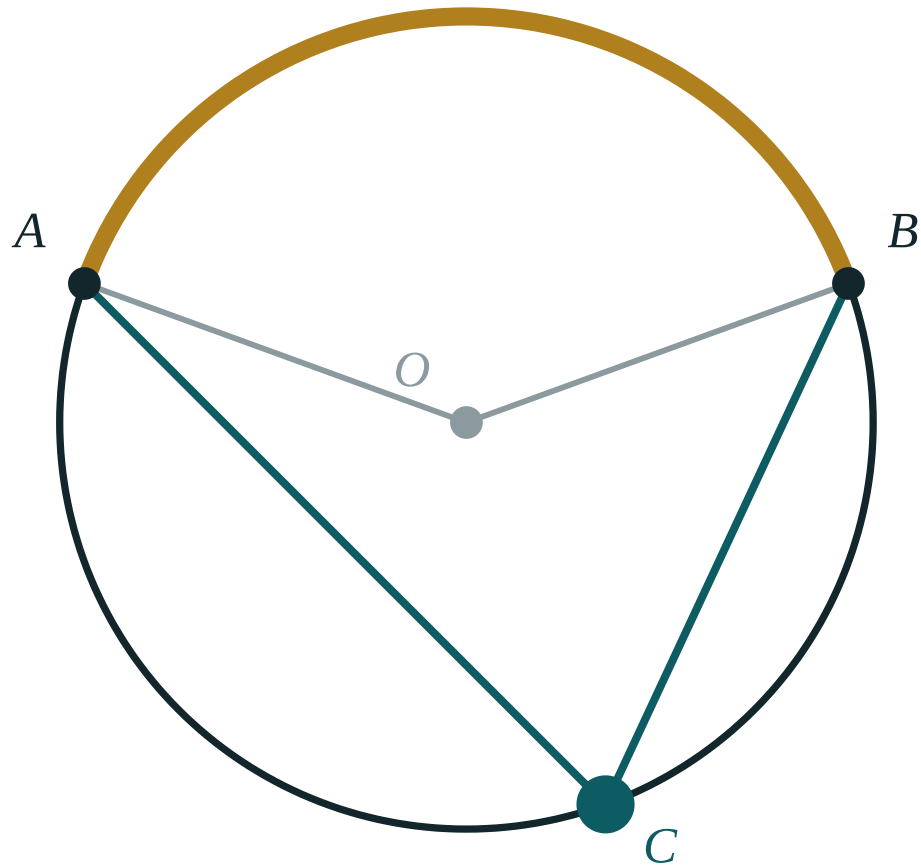
ANSWER

$$r = 2$$

03 Interactive model

Construct the three angle bisectors of a scalene triangle with compasses; the circle drawn from their meeting point touches all three sides at the first attempt.

The inscribed circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

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Circumscribed polygon	Tashqi chizilgan ko'pburchak	Описанный многоугольник
Inscribed circle	Ichki chizilgan aylana	Вписанная окружность
Incentre	Ichki markaz	Центр вписанной окружности
Inradius	Ichki radius	Радиус вписанной окружности
Angle bisector	Bissektrisa	Биссектриса
Tangent length	Urinma uzunligi	Длина касательной
Semiperimeter	Yarim perimetr	Полупериметр
Tangential quadrilateral	Tashqi chizilgan to'rtburchak	Описанный четырёхугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A circumscribed polygon has:

vertices on the circle

sides touching the circle

the circle outside

no circle

The incentre is where the _____ meet:

medians

angle bisectors

perpendicular bisectors

altitudes

The incentre is:

sometimes outside

always inside

on a side

at a vertex

S equals:

$$pr$$

$$2pr$$

$$\frac{pr}{2}$$

$$p + r$$

For 13, 14, 15, $r =$

$$2$$

$$3$$

$$4$$

$$6$$

A tangential quadrilateral satisfies:

$$a + b = c + d$$

$$a + c = b + d$$

$$a = c$$

$$abcd = 1$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3, 4, 5: p

ANSWER 6

2. Same: S

ANSWER 6

3. Same: r

ANSWER 1

4. 5, 12, 13: r

ANSWER 2

5. 13, 14, 15: r

ANSWER 4

6. Is a rhombus tangential?

ANSWER Yes

7. Is a rectangle (not a square) tangential?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Right triangle legs 6, 8: r

ANSWER 2

2. Right triangle legs 9, 12: r

ANSWER 3

3. Tangential quadrilateral 7, 9, 11, x

ANSWER $x = 9$

4. Tangential quadrilateral 5, 8, 10, x

ANSWER $x = 7$

5. A triangle with $S = 60$ and $p = 20$: r

ANSWER 3

6. An equilateral triangle of side 6: r

ANSWER $\sqrt{3}$

7. Which quadrilateral is both cyclic and tangential?

ANSWER The square

Hard · stretch and reason

7 PROBLEMS

1. A triangle 7, 8, 9: r

ANSWER ≈ 2.24

2. An equilateral triangle of side a : r

ANSWER $\frac{a}{2\sqrt{3}}$

3. Same: the ratio $R : r$

ANSWER $2 : 1$

4. A right triangle with legs a, b : r

ANSWER $\frac{a + b - c}{2}$

5. A rhombus of side 10 with diagonals 12 and 16: r

ANSWER 4.8

6. A tangential quadrilateral of perimeter 40: $a + c$

ANSWER 20

7. A triangle with $r = 3$ and perimeter 30: its area

ANSWER 45

07 Homework

Homework — 5 tasks

Write p as a fraction over 2 before using $S = pr$.

- Find the inradius of the triangle with sides 9, 10, 17.
- Find the inradius of a right triangle with legs 5 and 12.
- A tangential quadrilateral has sides 6, 11, 13, x in order. Find x .
- Find the inradius of an equilateral triangle of side 12.

5. Explain why a rectangle that is not a square has no inscribed circle touching all four sides.

Regular polygons

All sides equal and all angles equal — and one formula for every interior angle.

LESSON 37

1 × 40 MIN

GEOMETRIYA 9, §23

IGX 3.4

LESSON CLOCK

10'

14'

10'

6'

The definition and the angles	10 min
Working backwards	14 min
Tessellation	10 min
Homework	6 min

LEARNING OBJECTIVES

- Define a regular polygon and give the interior and exterior angle formulae.
- Find n from a given interior or exterior angle.
- Know which regular polygons tessellate.
- Compute the sum of the interior angles of any polygon.

TEXTBOOK

National	pp. 135–138
Cambridge	Core & Extended, pp. 58–63
Workbook	Exercise 3.4

01

Explanation

The definition and the angles

A polygon is **regular** if it is both equilateral and equiangular. Both conditions are needed: a rhombus is equilateral but not regular, a rectangle equiangular but not regular.

Sum of the interior angles: $(n - 2) \cdot 180^\circ$

Each interior angle: $\frac{(n - 2) \cdot 180^\circ}{n}$ Each exterior angle: $\frac{360^\circ}{n}$

<i>N</i>	NAME	INTERIOR	EXTERIOR	CENTRAL
3	triangle	60°	120°	120°
4	square	90°	90°	90°
5	pentagon	108°	72°	72°
6	hexagon	120°	60°	60°
8	octagon	135°	45°	45°
10	decagon	144°	36°	36°

THE EXTERIOR ANGLES ALWAYS TOTAL 360°

For every polygon, regular or not, convex or not. Walking once round the boundary turns you through one full circle — which is why the exterior angle formula has no $n - 2$ in it, and why it is the easier of the two to use.

Working backwards

GIVEN	FIND <i>N</i> BY	EXAMPLE
an exterior angle e	$n = \frac{360^\circ}{e}$	$e = 24^\circ \Rightarrow n = 15$
an interior angle i	$e = 180^\circ - i$, then divide	$i = 162^\circ \Rightarrow e = 18^\circ \Rightarrow n = 20$
the angle sum S	$n = \frac{S}{180^\circ} + 2$	$S = 1440^\circ \Rightarrow n = 10$

GO THROUGH THE EXTERIOR ANGLE

Solving $\frac{(n - 2)180}{n} = 162$ directly needs an equation; $180 - 162 = 18$ then $\frac{360}{18} = 20$ needs two subtractions. The second route is faster and far less error-prone.

Tessellation

A regular polygon **tessellates** — tiles the plane with no gaps — exactly when its interior angle divides 360° .

POLYGON	INTERIOR	$360^\circ \div$	TESSELLATES?
triangle	60°	6	yes
square	90°	4	yes
pentagon	108°	3.33...	no
hexagon	120°	3	yes
octagon	135°	2.67	no

Only three regular polygons tessellate on their own: the equilateral triangle, the square and the regular hexagon. The hexagon is the most efficient of the three, which is why bees build hexagonal cells.

02

Worked examples

EXAMPLE 1

A regular polygon has an interior angle of 162° . Find n .

1

Exterior angle = $180^\circ - 162^\circ = 18^\circ$.

2

$n = \frac{360^\circ}{18^\circ}$

3

$n = 20$

4

Check: $\frac{18 \cdot 180}{20} = 162 \checkmark$

ANSWER

$n = 20$

EXAMPLE 2 Find the sum of the interior angles of a polygon with 12 sides, and each angle if it is regular.

① $(12 - 2) \times 180^\circ = 1800^\circ$

② Each angle = $\frac{1800}{12}$

③ $= 150^\circ$

④ Check: exterior = $30^\circ = \frac{360}{12} \checkmark$

ANSWER

1800° ; each 150°

EXAMPLE 3 Does a regular pentagon tessellate?

① Its interior angle is 108° .

② $\frac{360}{108} = 3.33\dots$

Not a whole number.

③ Three pentagons leave a gap of 36° .

④ No.

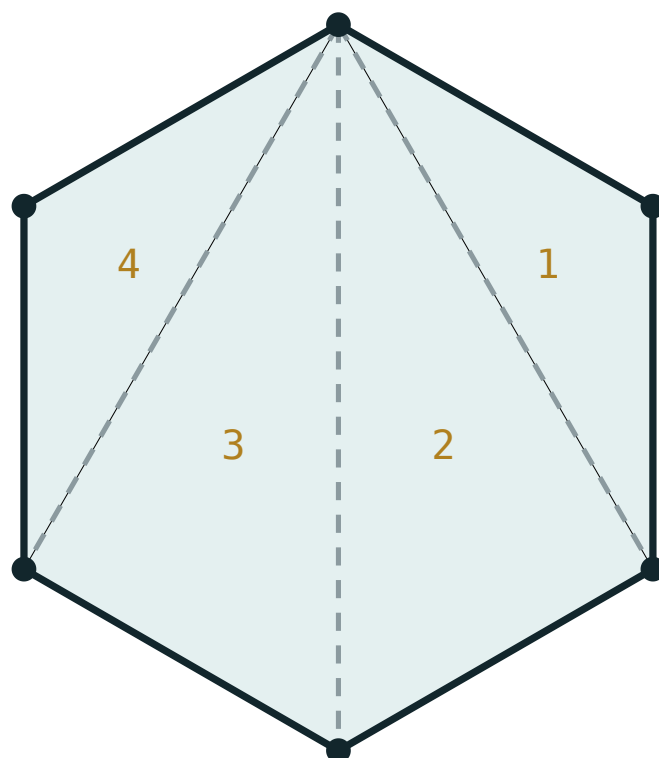
ANSWER

No

03 Interactive model

Give out card triangles, squares, pentagons and hexagons and let the class try to tile a desk; the pentagon fails visibly, and the arithmetic afterwards explains it.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**

triangles **4**

interior sum **720°**

each angle **120°**

exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Regular polygon	Muntazam ko'pburchak	Правильный многоугольник
Interior angle	Ichki burchak	Внутренний угол
Exterior angle	Tashqi burchak	Внешний угол
Angle sum	Burchaklar yig'indisi	Сумма углов
Central angle	Markaziy burchak	Центральный угол
Tessellation	Parketlash	Замошение
Equilateral	Teng tomonli	Равносторонний
Equiangular	Teng burchakli	Равноугольный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The interior angle sum is:

The exterior angles total:

A regular pentagon's interior angle:

An exterior angle of 24° gives:

$$n = 12$$

$$n = 15$$

$$n = 18$$

$$n = 20$$

An interior angle of 162° gives:

$$n = 18$$

$$n = 20$$

$$n = 24$$

$$n = 30$$

Which regular polygons tessellate?

all

triangle, square, hexagon

square only

pentagon and hexagon

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Interior angle of a regular hexagon

ANSWER 120°

2. Exterior angle of a regular hexagon

ANSWER 60°

3. Interior angle of a regular octagon

ANSWER 135°

4. Angle sum of a pentagon

ANSWER 540°

5. Angle sum of a decagon

ANSWER 1440°

6. Exterior angles always total

ANSWER 360°

7. Interior angle of a regular pentagon

ANSWER 108°

Medium · the standard question

7 PROBLEMS

1. Exterior angle 24° : n

ANSWER 15

2. Interior angle 162° : n

ANSWER 20

3. Angle sum 1800° : n

ANSWER 12

4. Interior angle of a regular 12-gon

ANSWER 150°

5. Interior angle of a regular 9-gon

ANSWER 140°

6. Does a regular pentagon tessellate?

ANSWER No

7. Does a regular hexagon tessellate?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. A regular polygon whose interior angle is 5 times its exterior

ANSWER $n = 12$

2. A regular polygon with n diagonals from one vertex is a

ANSWER $(n + 3)$ -gon

3. Total diagonals of a regular decagon

ANSWER 35

4. A regular polygon with interior angle 175°

ANSWER $n = 72$

5. Two regular polygons tile with a square at a point: which?

ANSWER Two octagons and a square

6. Interior angle of a regular 20-gon

ANSWER 162°

7. Is there a regular polygon with interior angle 155° ?

ANSWER No — $\frac{360}{25}$ is not whole

07 Homework

Homework — 5 tasks

Find n through the exterior angle, never by solving the interior-angle equation.

- Find the interior and exterior angles of a regular 15-gon.
- A regular polygon has an exterior angle of 20° . Find n and the angle sum.
- A regular polygon has an interior angle of 156° . Find n .
- Which regular polygons tessellate, and why?

5. How many diagonals has a regular 12-gon?

Circles inscribed in and circumscribed about a regular polygon

Every regular polygon has both circles, and they share a centre.

LESSON 38–39

2 × 40 MIN

GEOMETRIYA 9, §24

IGX 3.4–3.5

LESSON CLOCK

13'

23'

22'

18'

4'

One centre, two circles	13 min
The central triangle	23 min
The area	22 min
The standard polygons	18 min
Homework	4 min

LEARNING OBJECTIVES

- Prove that a regular polygon has both an inscribed and a circumscribed circle with the same centre.
- Use the central triangle to relate R , r and the side a .
- Compute the area of a regular polygon as $\frac{1}{2} P r$.
- Find R and r for the standard polygons.

TEXTBOOK

National	pp. 139–145
Cambridge	Core & Extended, pp. 58–70
Workbook	Exercise 3.4

01

Explanation

One centre, two circles

Join the centre O of a regular n -gon to all n vertices. The n triangles formed are congruent isosceles triangles, so:

ALL EQUAL	MEANING
the distances O to the vertices	a circumscribed circle of radius R
the distances O to the sides	an inscribed circle of radius r

The inradius r is also called the **apothem**: it is the distance from the centre to the midpoint of a side, and it is the height of one of the n triangles.

ONLY REGULAR POLYGONS HAVE BOTH CIRCLES CONCENTRIC

A general triangle has both circles, but with different centres. A rhombus has only the inscribed one, a rectangle only the circumscribed one. Regularity is exactly the condition that makes the two centres coincide.

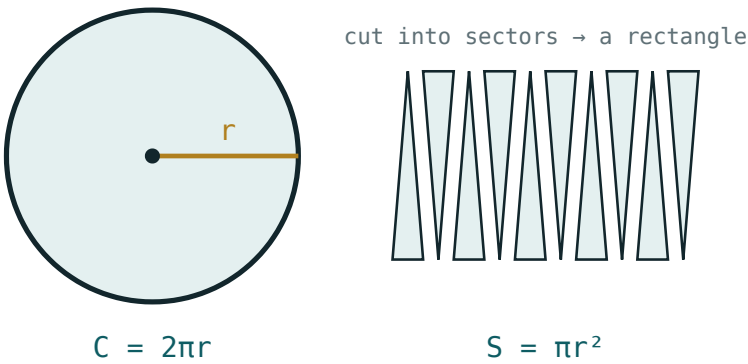
The central triangle

One of the n triangles has apex angle $\frac{360^\circ}{n}$ at O , two sides R , base a and height r .

Halving it gives a right triangle with acute angle $\frac{180^\circ}{n}$:

$$\frac{a}{2} = R \sin \frac{180^\circ}{n} \quad r = R \cos \frac{180^\circ}{n}$$

$$a = 2R \sin \frac{180^\circ}{n} \quad a = 2r \tan \frac{180^\circ}{n}$$



One of the n congruent triangles — its height is the apothem, its base a side.

KNOWN	FIND	FORMULA
R, n	a	$2R \sin \frac{180^\circ}{n}$
a, n	R	$\frac{a}{2 \sin (180^\circ/n)}$
a, n	r	$\frac{a}{2 \tan (180^\circ/n)}$
R, n	r	$R \cos \frac{180^\circ}{n}$

THE ANGLE IN THE FORMULA IS $\frac{180^\circ}{N}$, NOT $\frac{360^\circ}{N}$

Because the right triangle is **half** of the central one. Using the full central angle doubles every side — the commonest error of this lesson.

The area

The polygon is n triangles, each of area $\frac{1}{2}ar$:

$$S = n \cdot \frac{1}{2}ar = \frac{1}{2}Pr$$

where $P = na$ is the perimeter. This is exactly the formula $S = pr$ of the last lesson, with $p = \frac{P}{2}$.

POLYGON OF SIDE A	AREA
equilateral triangle	$\frac{a^2\sqrt{3}}{4}$
square	a^2
regular hexagon	$\frac{3a^2\sqrt{3}}{2}$

A REGULAR HEXAGON IS SIX EQUILATERAL TRIANGLES

Its central angle is 60° , so each of the six triangles is equilateral and $R = a$. That single fact answers most hexagon questions in one line.

The standard polygons

POLYGON	R	r	A
equilateral triangle	$\frac{a}{\sqrt{3}}$	$\frac{a}{2\sqrt{3}}$	$R\sqrt{3}$
square	$\frac{a\sqrt{2}}{2}$	$\frac{a}{2}$	$R\sqrt{2}$
regular hexagon	a	$\frac{a\sqrt{3}}{2}$	R

These three are worth knowing by heart: the ratio $R : r$ is $2 : 1$ for the triangle, $\sqrt{2} : 1$ for the square and $2 : \sqrt{3}$ for the hexagon.

$R = A$ IS TRUE ONLY FOR THE HEXAGON

For the square R is larger than half the side by a factor of $\sqrt{2}$; for the triangle larger still. Assuming $R = a$ in general is a serious error.

02 Worked examples

EXAMPLE 1 A regular hexagon has side 6. Find R , r and its area.

① Central angle 60° — the triangles are equilateral, so $R = 6$.

② $r = R \cos 30^\circ = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$

③ $P = 36$

④ $S = \frac{1}{2} \times 36 \times 3\sqrt{3} = 54\sqrt{3} \approx 93.5$

ANSWER

$$R = 6, r = 3\sqrt{3}, S = 54\sqrt{3}$$

EXAMPLE 2 A regular pentagon has circumradius 10. Find its side and area.

① $a = 2R \sin 36^\circ = 20 \times 0.5878$

② $a \approx 11.76$

③ $r = R \cos 36^\circ = 10 \times 0.8090 \approx 8.09$

④ $S = \frac{1}{2} \times 5 \times 11.76 \times 8.09 \approx 238$

ANSWER

$$a \approx 11.8, S \approx 238$$

EXAMPLE 3 A square has side 8. Find R and r .

$$① \quad r = \frac{a}{2} = 4$$

Half the side.

$$② \quad R = \frac{a\sqrt{2}}{2} = 4\sqrt{2}$$

Half the diagonal.

$$③ \quad R \approx 5.66$$

$$④ \quad R : r = \sqrt{2} : 1 \quad \checkmark$$

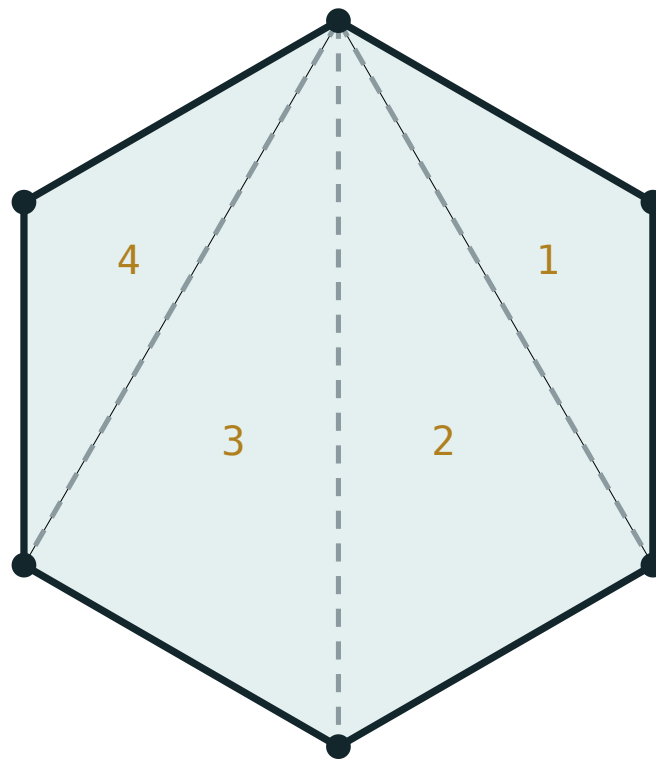
ANSWER

$$R = 4\sqrt{2}, r = 4$$

03 Interactive model

Draw a regular hexagon by stepping the radius round a circle with compasses; the six equal chords prove $R = a$ without any algebra.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**triangles **4**interior sum **720°**each angle **120°**exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Centre of a regular polygon	Muntazam ko'pburchak markazi	Центр правильного многоугольника
Apothem	Apofema	Апофема
Central triangle	Markaziy uchburchak	Центральный треугольник
Circumradius	Tashqi radius	Радиус описанной окружности
Inradius	Ichki radius	Радиус вписанной окружности
Half-angle	Yarim burchak	Половинный угол
Perimeter	Perimetr	Периметр
Area of a polygon	Ko'pburchak yuzi	Площадь многоугольника

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

A regular polygon has:

The apothem is:

a equals:

The area of a regular polygon is:

$$Pr$$

$$\frac{1}{2}Pr$$

$$2Pr$$

$$Pr^2$$

For a regular hexagon:

$$R = 2a$$

$$R = a$$

$$R = \frac{a}{2}$$

$$R = a\sqrt{3}$$

For a square, $R : r$ is:

$$2 : 1$$

$$\sqrt{2} : 1$$

$$1 : 1$$

$$2 : \sqrt{3}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Hexagon of side 6: R

ANSWER 6

2. Hexagon of side 6: r

ANSWER $3\sqrt{3}$

3. Square of side 8: r

ANSWER 4

4. Square of side 8: R

ANSWER $4\sqrt{2}$

5. Equilateral triangle of side 6: R

ANSWER $2\sqrt{3}$

6. Same: r

ANSWER $\sqrt{3}$

7. Area of a regular polygon

ANSWER $\frac{1}{2}Pr$

Medium · the standard question

7 PROBLEMS

1. Hexagon of side 6: its area

ANSWER $54\sqrt{3}$

2. Square of side 8: its area

ANSWER 64

3. Pentagon with $R = 10$: its side

ANSWER ≈ 11.8

4. Same: its area

ANSWER ≈ 238

5. Octagon of side 4: its apothem

ANSWER ≈ 4.83

6. Same: its area

ANSWER ≈ 77.3

7. Hexagon with $R = 10$: its area

ANSWER $150\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. A regular 12-gon with $R = 6$: its area

ANSWER 108

2. A regular polygon with $a = 5$ and $r = 6$: n

ANSWER 8

3. A hexagon inscribed in a circle of radius 8: its perimeter

ANSWER 48

4. A square inscribed in a circle of radius 8: its area

ANSWER 128

5. An equilateral triangle inscribed in a circle of radius 8: its side

ANSWER $8\sqrt{3}$

6. The ratio of the areas of the inscribed square and hexagon of the same circle

ANSWER $4 : 3\sqrt{3}$

7. As $n \rightarrow \infty$, $\frac{r}{R}$ tends to

ANSWER 1

07 Homework

Homework — 5 tasks

Draw the central triangle and mark the half-angle $\frac{180^\circ}{n}$ every time.

1. A regular hexagon has side 10. Find R , r and its area.
2. A square has circumradius 6. Find its side and area.
3. A regular octagon has side 5. Find its apothem and area.
4. An equilateral triangle is inscribed in a circle of radius 9. Find its side.

5. Explain why $R = a$ for a regular hexagon but not for a square.

The side of a regular polygon and the radii

Exact expressions for the triangle, the square and the hexagon — and one recurrence for doubling n .

LESSON 40–41

2 × 40 MIN

GEOMETRIYA 9, §25

IGX 3.4

LESSON CLOCK

13'

23'

22'

18'

4'

Three exact sides	13 min
The general formula	23 min
Doubling n	22 min
Towards the circumference	18 min
Homework	4 min

LEARNING OBJECTIVES

- Derive the exact side of the inscribed triangle, square and hexagon in terms of R .
- Use $a = 2R \sin(180^\circ/n)$ for any n .
- Use the doubling formula to pass from an n -gon to a $2n$ -gon.
- Compare the perimeters of inscribed polygons with the circumference.

TEXTBOOK

National	pp. 146–152
Cambridge	Core & Extended, pp. 58–63
Workbook	Exercise 3.4

01

Explanation

Three exact sides

For a polygon inscribed in a circle of radius R , the central triangle gives the side directly.

N	CENTRAL ANGLE	$A = 2R \sin \frac{180^\circ}{N}$	EXACT
3	120°	$2R \sin 60^\circ$	$R\sqrt{3}$
4	90°	$2R \sin 45^\circ$	$R\sqrt{2}$
6	60°	$2R \sin 30^\circ$	R
8	45°	$2R \sin 22.5^\circ$	$R\sqrt{2 - \sqrt{2}}$
12	30°	$2R \sin 15^\circ$	$R\sqrt{2 - \sqrt{3}}$

THREE THAT MUST BE AUTOMATIC

$a_3 = R\sqrt{3}, a_4 = R\sqrt{2}, a_6 = R$. The pattern $\sqrt{3}, \sqrt{2}, \sqrt{1}$ is easy to hold, and these three cover almost every examination question.

The general formula

$$a_n = 2R \sin \frac{180^\circ}{n} \quad r_n = R \cos \frac{180^\circ}{n}$$

GIVEN	FIND	FORMULA
R, n	a	$2R \sin \frac{180^\circ}{n}$
a, n	R	$\frac{a}{2 \sin (180^\circ/n)}$
r, n	R	$\frac{r}{\cos (180^\circ/n)}$
R, r	a	$2\sqrt{R^2 - r^2}$

The last row is Pythagoras in the half-central triangle, and it needs no trigonometry at all — useful when neither n nor a calculator is available.

$$R^2 = R^2 + \frac{A^2}{4}$$

This one relation ties all three quantities together. Given any two of R, r, a , it gives the third without knowing n .

Doubling n

From the side of an inscribed n -gon, the side of the $2n$ -gon follows:

$$a_{2n} = \sqrt{2R^2 - 2R\sqrt{R^2 - \frac{a_n^2}{4}}}$$

Starting from the hexagon ($a_6 = R$) and doubling repeatedly gives the 12-gon, the 24-gon, the 48-gon — which is how Archimedes computed π to two decimal places, about 250 BC.

N	$A_n \div R$	PERIMETER $\div 2R$
6	1	3.000
12	0.5176	3.106
24	0.2611	3.133
48	0.1308	3.139
96	0.0654	3.141

THE LAST COLUMN IS π APPEARING

The perimeter of an inscribed n -gon divided by the diameter creeps up towards 3.14159.... The next lesson gives that limit its name.

Towards the circumference

As n grows, three things happen together:

QUANTITY	BEHAVIOUR	LIMIT
a_n	decreases	0
r_n	increases	R
the perimeter na_n	increases	the circumference
the area	increases	the area of the disc

The inscribed polygon is always **smaller** than the circle and the circumscribed one always larger, so the circumference is trapped between two computable numbers —

which is exactly how Archimedes bounded π between $\frac{223}{71}$ and $\frac{22}{7}$.

A POLYGON NEVER BECOMES A CIRCLE

For every finite n the polygon has corners and the circle does not. The circle is the **limit**, a word that will be defined properly in Grade 11 — but the idea is already clear here.

02 Worked examples

EXAMPLE 1 A regular hexagon is inscribed in a circle of radius 8. Find its side and perimeter.

$$① \quad a_6 = 2R \sin 30^\circ$$

$$② \quad = 2 \times 8 \times 0.5$$

$$③ \quad a = 8$$

Equal to R .

$$④ \quad P = 48$$

ANSWER

$$a = 8, P = 48$$

EXAMPLE 2 A square is inscribed in a circle of radius 5. Find its side, apothem and area.

$$① \quad a = R\sqrt{2} = 5\sqrt{2} \approx 7.07$$

$$② \quad r = R \cos 45^\circ = \frac{5\sqrt{2}}{2} \approx 3.54$$

$$③ \quad S = a^2 = 50$$

$$④ \quad \text{Check: } \frac{1}{2} \times 4 \times 7.07 \times 3.54 \approx 50 \quad \checkmark$$

ANSWER

$$a = 5\sqrt{2}, r = \frac{5\sqrt{2}}{2}, S = 50$$

EXAMPLE 3 A regular polygon inscribed in a circle of radius 10 has apothem 8. Find its side.

$$1 \quad R^2 = r^2 + \frac{a^2}{4}$$

$$2 \quad 100 = 64 + \frac{a^2}{4}$$

$$3 \quad \frac{a^2}{4} = 36 \Rightarrow a^2 = 144$$

$$4 \quad a = 12$$

No trigonometry needed.

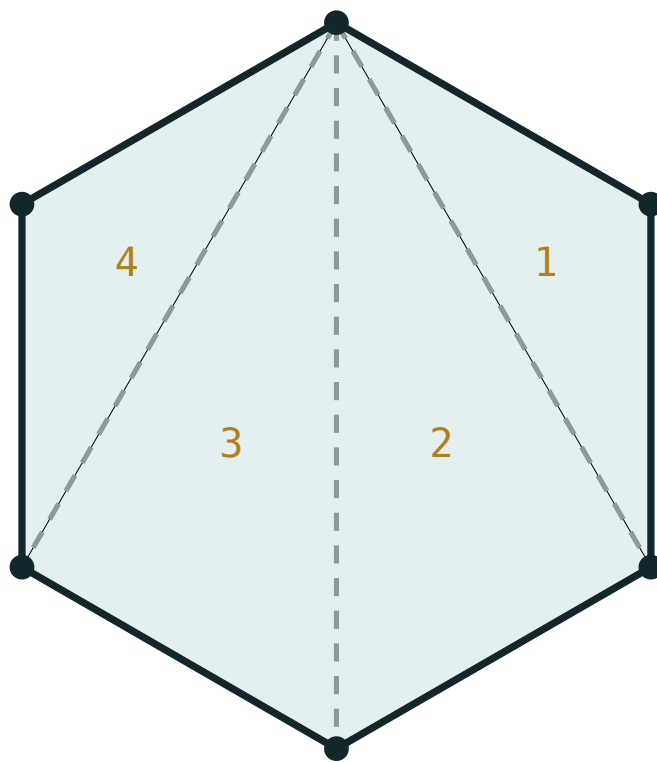
ANSWER

$$a = 12$$

03 Interactive model

Draw a hexagon, a 12-gon and a 24-gon in one circle on the board; the 24-gon is already visually a circle, and the class sees the limit forming.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**triangles **4**interior sum **720°**each angle **120°**exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Side of a polygon	Ko'pburchak tomoni	Сторона многоугольника
Inscribed in a circle	Aylanaga ichki chizilgan	Вписанный в окружность
Exact expression	Aniq ifoda	Точное выражение
Doubling formula	Ikkilantirish formulasi	Формула удвоения
Approximation	Taqribiy qiymat	Приближение
Perimeter	Perimetr	Периметр
Limit	Limit	Предел
Archimedes	Arximed	Архимед

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** a_6 equals:

R

$R\sqrt{2}$

$R\sqrt{3}$

$2R$

 a_4 equals:

R

$R\sqrt{2}$

$R\sqrt{3}$

$2R$

 a_3 equals:

R

$R\sqrt{2}$

$R\sqrt{3}$

$3R$

R^2 equals:

$$r^2 + a^2$$

$$r^2 + \frac{a^2}{4}$$

$$r^2 - a^2$$

$$2r^2$$

As n grows, r_n :

decreases to 0

increases to R

stays fixed

oscillates

The inscribed perimeter is:

greater than the circumference

less than it

equal to it

unrelated

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. a_6 for $R = 8$

ANSWER 8

2. a_4 for $R = 5$

ANSWER $5\sqrt{2}$

3. a_3 for $R = 6$

ANSWER $6\sqrt{3}$

4. r_6 for $R = 8$

ANSWER $4\sqrt{3}$

5. r_4 for $R = 5$

ANSWER $\frac{5\sqrt{2}}{2}$

6. Perimeter of a hexagon in $R = 8$

ANSWER 48

7. Area of a square in $R = 5$

ANSWER 50

Medium · the standard question

7 PROBLEMS

1. $R = 10, r = 8: a$

ANSWER 12

2. $R = 13, a = 24: r$

ANSWER 5

3. a_{12} for $R = 10$

ANSWER ≈ 5.18

4. Perimeter of a 12-gon in $R = 10$

ANSWER ≈ 62.1

5. Circumference of that circle

ANSWER ≈ 62.8

6. a_8 for $R = 4$

ANSWER ≈ 3.06

7. Area of a hexagon in $R = 8$

ANSWER $96\sqrt{3}$
Hard · stretch and reason

7 PROBLEMS

1. A regular 24-gon in $R = 1$: its perimeter

ANSWER ≈ 6.265

2. Compare with 2π

ANSWER ≈ 6.283 — the polygon is smaller

3. A hexagon and a square in the same circle: the ratio of their perimeters

ANSWER $6R : 4R\sqrt{2}$, i.e. $3 : 2\sqrt{2}$

4. A regular polygon with $a = R$: n

ANSWER 6

5. A regular polygon with $a = R\sqrt{2}$: n

ANSWER 4

6. A regular n -gon has $r = \frac{R}{2}$: find n

ANSWER 3

7. Archimedes bounded π between

ANSWER $\frac{223}{71}$ and $\frac{22}{7}$

07 Homework

Homework — 5 tasks

Use $R^2 = r^2 + \frac{a^2}{4}$ whenever two of the three are known.

1. A regular hexagon is inscribed in a circle of radius 12. Find its side, apothem and area.
2. A square is inscribed in a circle of radius 9. Find its side and area.
3. An equilateral triangle is inscribed in a circle of radius 10. Find its side.
4. A regular polygon inscribed in a circle of radius 13 has side 10. Find its apothem.

5. Compare the perimeter of a regular 12-gon inscribed in a unit circle with 2π .

The circumference of a circle

One number, the same for every circle — and the formula that follows from it.

LESSON 42–43

2 × 40 MIN

GEOMETRIYA 9, §26

IGX 7.1

LESSON CLOCK

13'

23'

22'

18'

4'

What π is	13 min
The formula	23 min
Wheels and belts	22 min
Composite perimeters	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define π as the ratio of the circumference to the diameter.
- Use $C = 2\pi R = \pi d$ in both directions.
- Solve problems on wheels, belts and perimeters of composite figures.
- Give answers exactly in terms of π and as decimals.

TEXTBOOK

National	pp. 153–158
Cambridge	Core & Extended, pp. 130–136
Workbook	Exercise 7.1

01

Explanation

What π is

Measure the circumference C and the diameter d of any circle and divide. The answer is the same every time:

$$\pi = \frac{C}{d} = 3.14159265\dots$$

This is not a coincidence but a consequence of the fact that all circles are similar: a circle of twice the radius is an enlargement with $k = 2$, so its circumference is twice as long.

APPROXIMATION	VALUE	ERROR
3	3	$\approx 4.5\%$
$\frac{22}{7}$	3.1428...	$\approx 0.04\%$
3.14	3.14	$\approx 0.05\%$
$\frac{355}{113}$	3.14159292...	$\approx 0.000009\%$

π IS IRRATIONAL – AND THAT WAS PROVED ONLY IN 1761

No fraction equals it exactly, and no decimal expansion terminates or repeats. Every value above is an approximation, which is why exact answers are left in terms of π .

The formula

$C = \pi d = 2\pi R$

GIVEN	FIND	WORKING
$R = 7$	C	$14\pi \approx 44.0$
$d = 10$	C	$10\pi \approx 31.4$
$C = 62.8$	R	$\frac{62.8}{2\pi} \approx 10$
$C = 12\pi$	d	12

EXACT MEANS “IN TERMS OF π ”

A circle of radius 7 has circumference exactly 14π ; 43.98 is a rounded value. When a question says “give your answer in terms of π ”, a decimal scores nothing.

Wheels and belts

One revolution of a wheel moves it forward by one circumference.

QUESTION	WORKING
a wheel of radius 30 cm in 100 turns	$100 \times 60\pi \approx 188\text{ m}$
turns needed to travel 1 km	$\frac{100000}{60\pi} \approx 531$
a belt round two equal pulleys of radius r , centres d apart	$2\pi r + 2d$

The last row is worth thinking through: the two straight parts total $2d$, and the two arcs together make one complete circle.

DISTANCE $\div$ CIRCUMFERENCE = NUMBER OF TURNS

Every wheel question is one of those two divisions. Deciding which before starting takes five seconds and settles the whole problem.

Composite perimeters

FIGURE	PERIMETER
a semicircle of radius R	$\pi R + 2R$
a quarter-circle of radius R	$\frac{\pi R}{2} + 2R$
a running track: two straights L , two semicircular ends of radius R	$2L + 2\pi R$
a square of side a with a semicircle on one side	$3a + \frac{\pi a}{2}$

A SEMICIRCLE'S PERIMETER INCLUDES THE DIAMETER

πR is only the curved part. The boundary of the region also runs along the straight edge, so $2R$ must be added. Omitting it is the single most common error in composite-perimeter questions.

02 Worked examples

EXAMPLE 1 A circle has radius 7 cm. Find its circumference exactly and to 3 s.f.

① $C = 2\pi R = 14\pi$

Exact.

② $14 \times 3.14159\dots$

③ ≈ 43.98

④ $\approx 44.0 \text{ cm}$

Three significant figures.

ANSWER

$$14\pi \approx 44.0 \text{ cm}$$

EXAMPLE 2 A bicycle wheel has radius 30 cm. How many turns to travel 1 km?

① $C = 60\pi \approx 188.5 \text{ cm}$

② $1 \text{ km} = 100\,000 \text{ cm}$

③
$$\frac{100000}{188.5}$$

④ $\approx 531 \text{ turns.}$

ANSWER

$$\approx 531$$

EXAMPLE 3 Find the perimeter of a semicircular region of radius 10.

① Curved part: $\frac{1}{2} \times 2\pi \times 10 = 10\pi$.

② Straight part: the diameter, 20.

③ $P = 10\pi + 20$

④ ≈ 51.4

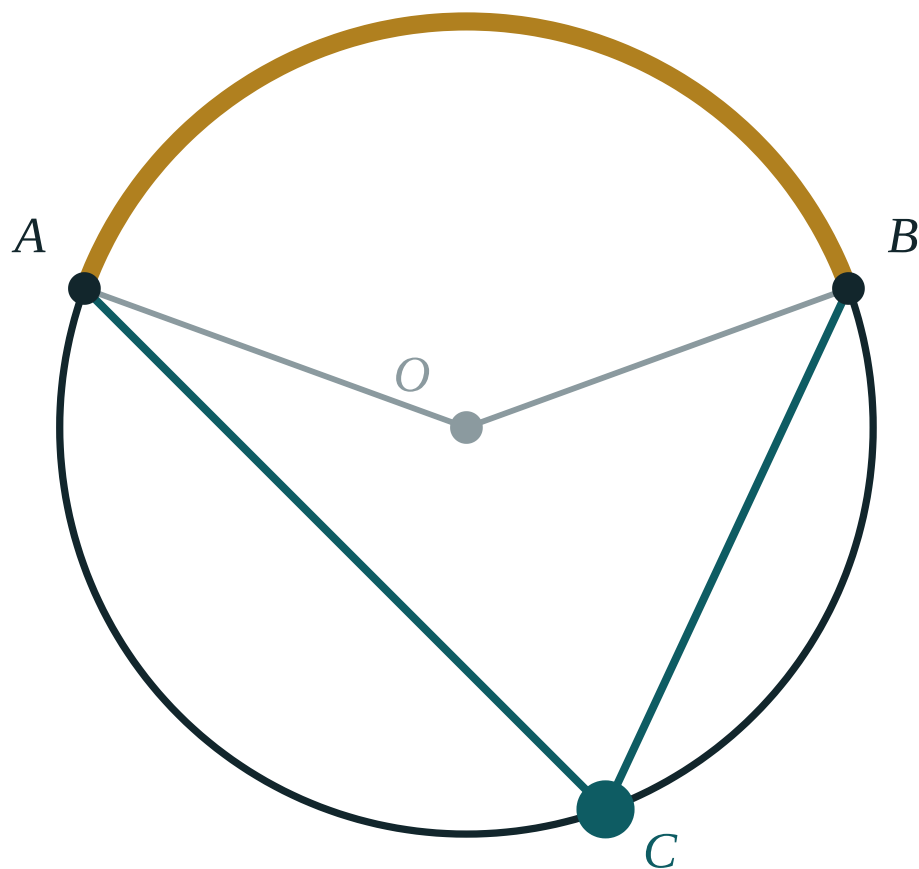
ANSWER

$$10\pi + 20 \approx 51.4$$

03 Interactive model

Measure a tin lid with string and a ruler, divide, and collect the whole class's answers on the board; they cluster around 3.14 without anyone being told to expect it.

C and d together



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Circumference	Aylana uzunligi	Длина окружности
Diameter	Diametr	Диаметр
Radius	Radius	Радиус
Pi	Pi soni	Число пи
Irrational	Irratsional	Иррациональное
Composite figure	Murakkab shakl	Составная фигура
Revolution	Aylanish	Оборот
Exactly	Aniq	Точно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** π is:

$$\frac{C}{R}$$

$$\frac{C}{d}$$

$$\frac{d}{C}$$

$$Cd$$

 C equals:

$$\pi R$$

$$2\pi R$$

$$\pi R^2$$

$$2\pi R^2$$

 $R = 7$ gives $C =$

$$7\pi$$

$$14\pi$$

$$49\pi$$

$$21\pi$$

$C = 12\pi$ gives $d =$

6

12

24

36

The perimeter of a semicircle of radius R :

πR

$\pi R + 2R$

$2\pi R$

$\pi R + R$

Turns to travel a distance D :

DC

$\frac{D}{C}$

$\frac{C}{D}$

$D + C$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $R = 7$: C exactly

ANSWER 14π

2. $R = 5$: C exactly

ANSWER 10π

3. $d = 10$: C exactly

ANSWER 10π

4. $C = 12\pi$: R

ANSWER 6

5. $C = 20\pi$: d

ANSWER 20

6. $R = 7$: C to 3 s.f.

ANSWER 44.0

7. Perimeter of a semicircle, $R = 10$

ANSWER $10\pi + 20$

Medium · the standard question

7 PROBLEMS

1. A wheel $R = 30\text{ cm}$: distance in 100 turns

ANSWER $\approx 188\text{ m}$

2. Same wheel: turns to travel 1 km

ANSWER ≈ 531

3. Perimeter of a quarter-circle, $R = 8$

ANSWER $4\pi + 16$

4. A track: straights 100 m , ends of radius 35 m

ANSWER $200 + 70\pi \approx 420\text{ m}$

5. A belt round two pulleys of radius 5, centres 20 apart

ANSWER $10\pi + 40$

6. $C = 31.4$: R

ANSWER ≈ 5

7. A square of side 6 with a semicircle on one side: its perimeter

ANSWER $18 + 3\pi$

Hard · stretch and reason

7 PROBLEMS

1. A circle whose circumference equals its diameter times π : always true?

ANSWER Yes — that is the definition

2. Two circles of radii 3 and 5: the difference of their circumferences

ANSWER 4π

3. A circle of circumference C : the circumference when R is doubled

ANSWER $2C$

4. Earth's equator, $R \approx 6371 \text{ km}$: its length

ANSWER $\approx 40\,030 \text{ km}$

5. A rope round the equator, raised 1 m everywhere: extra length needed

ANSWER $2\pi \approx 6.28 \text{ m}$

6. A running track lane 1.22 m wide: the extra length of the outer lane

ANSWER $\approx 7.67 \text{ m}$

7. Three equal circles of radius r touching pairwise: the perimeter of the curved triangle between

ANSWER πr

07 Homework

Homework — 5 tasks

Give every answer twice: exactly in terms of π , and to three significant figures.

- Find the circumference of a circle of radius 9 cm.
- A circle has circumference 44 cm. Find its radius, taking $\pi = \frac{22}{7}$.
- A wheel of diameter 70 cm makes 500 turns. How far does it travel?
- Find the perimeter of a semicircular region of radius 14.

5. A rope stretched round the Earth is raised 1 m at every point. How much extra rope is needed?

The length of an arc, and the radian measure of an angle

A fraction of a circle, and the unit of angle that makes the fraction disappear.

LESSON 44–45

2 × 40 MIN

GEOMETRIYA 9, §27

IGX 7.1

LESSON CLOCK

13'

23'

22'

18'

4'

An arc as a fraction	13 min
The radian	23 min
$l = R\alpha$	22 min
Sector perimeters	18 min
Homework	4 min

LEARNING OBJECTIVES

- Find an arc length as a fraction of the circumference.
- Convert between degrees and radians.
- Use $l = R\alpha$ with α in radians.
- Solve problems on arcs, sector perimeters and angular measure.

TEXTBOOK

National	pp. 159–165
Cambridge	Core & Extended, pp. 130–136
Workbook	Exercise 7.1

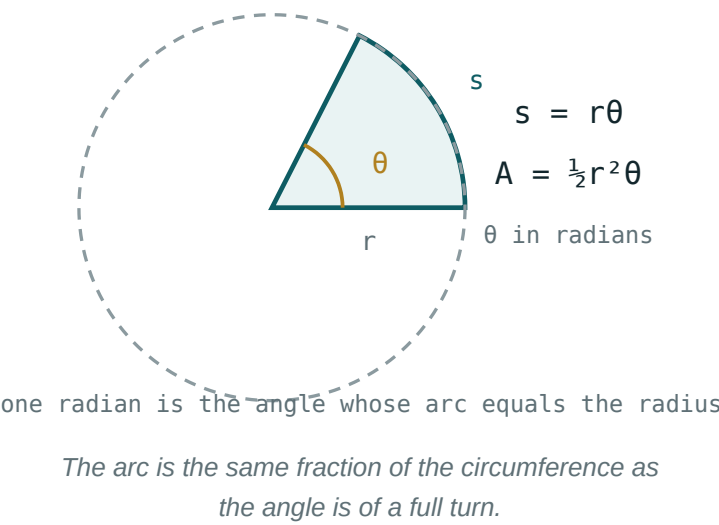
01

Explanation

An arc as a fraction

A central angle of θ degrees cuts off $\frac{\theta}{360}$ of the whole circle, so

$$l = \frac{\theta}{360^\circ} \cdot 2\pi R$$



θ	FRACTION	ARC, $R = 12$
90°	$\frac{1}{4}$	6π
60°	$\frac{1}{6}$	4π
45°	$\frac{1}{8}$	3π
120°	$\frac{1}{3}$	8π
30°	$\frac{1}{12}$	2π

EVERY ARC QUESTION IS A FRACTION QUESTION

Write the fraction first, then multiply. Doing the fraction and the circumference in one step is where the arithmetic errors live.

The radian

One **radian** is the central angle whose arc equals the radius. Since a full turn has arc $2\pi R$, a full turn is 2π radians:

$$180^\circ = \pi \text{ rad } 1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57.3^\circ$$

DEGREES	30°	45°	60°	90°	180°	360°
Radians	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

Degrees $\rightarrow$ radians: multiply by $\frac{\pi}{180^\circ}$. Radians $\rightarrow$ degrees: multiply by $\frac{180^\circ}{\pi}$.

THE SAME DEFINITION AS IN THE ALGEBRA COURSE

Grade 9 algebra introduced the radian from the unit circle in Quarter II; geometry meets it here through the arc. Both are the same statement, $\alpha = \frac{l}{R}$, and the two courses agree exactly.

$l = R\alpha$

With α in radians the fraction $\frac{\theta}{360}$ disappears entirely:

$$\alpha = \frac{l}{R} \implies l = R\alpha$$

GIVEN	FIND	WORKING
$R = 12, \alpha = \frac{\pi}{3}$	l	4π
$R = 5, \alpha = 2$	l	10
$l = 15, \alpha = 3$	R	5
$l = 6\pi, R = 12$	α	$\frac{\pi}{2}$

$L = RA$ IS FALSE IN DEGREES

With $\alpha = 60$ meaning degrees, $R\alpha$ would be 720 for $R = 12$ — absurd. Convert first, or use the $\frac{\theta}{360}$ form. Mixing the two is the standard error.

Sector perimeters

The boundary of a sector is one arc and two radii:

$$P = l + 2R = R\alpha + 2R = R(\alpha + 2)$$

SECTOR	PERIMETER
$R = 6, \alpha = \frac{\pi}{3}$	$2\pi + 12$
$R = 10, \theta = 90^\circ$	$5\pi + 20$
$a \text{ semicircle}, R$	$\pi R + 2R$
a segment cut by a chord	arc + chord

A SECTOR HAS TWO RADII, A SEGMENT HAS A CHORD

They are different regions with different boundaries. Reading which one a question means — and drawing it — settles the perimeter before any arithmetic.

02

Worked examples

EXAMPLE 1 Find the length of an arc of 60° in a circle of radius 12.

1

$$\frac{60}{360} = \frac{1}{6}$$

The fraction.

2

$$C = 24\pi$$

3

$$l = \frac{1}{6} \times 24\pi$$

4

$$= 4\pi \approx 12.6$$

ANSWER

$$4\pi \approx 12.6$$

EXAMPLE 2 Convert 135° to radians and $\frac{5\pi}{6}$ to degrees.

$$① \quad 135 \times \frac{\pi}{180} = \frac{135\pi}{180}$$

$$② \quad \text{Cancel by 45: } \frac{3\pi}{4}.$$

$$③ \quad \frac{5\pi}{6} \times \frac{180}{\pi} = \frac{900}{6}$$

$$④ \quad = 150^\circ$$

ANSWER

$$\frac{3\pi}{4} \text{ and } 150^\circ$$

EXAMPLE 3 A sector has radius 10 and angle 90° . Find its perimeter.

$$① \quad \alpha = \frac{\pi}{2}$$

$$② \quad l = 10 \times \frac{\pi}{2} = 5\pi$$

$$③ \quad \text{Two radii: } 20.$$

$$④ \quad P = 5\pi + 20 \approx 35.7$$

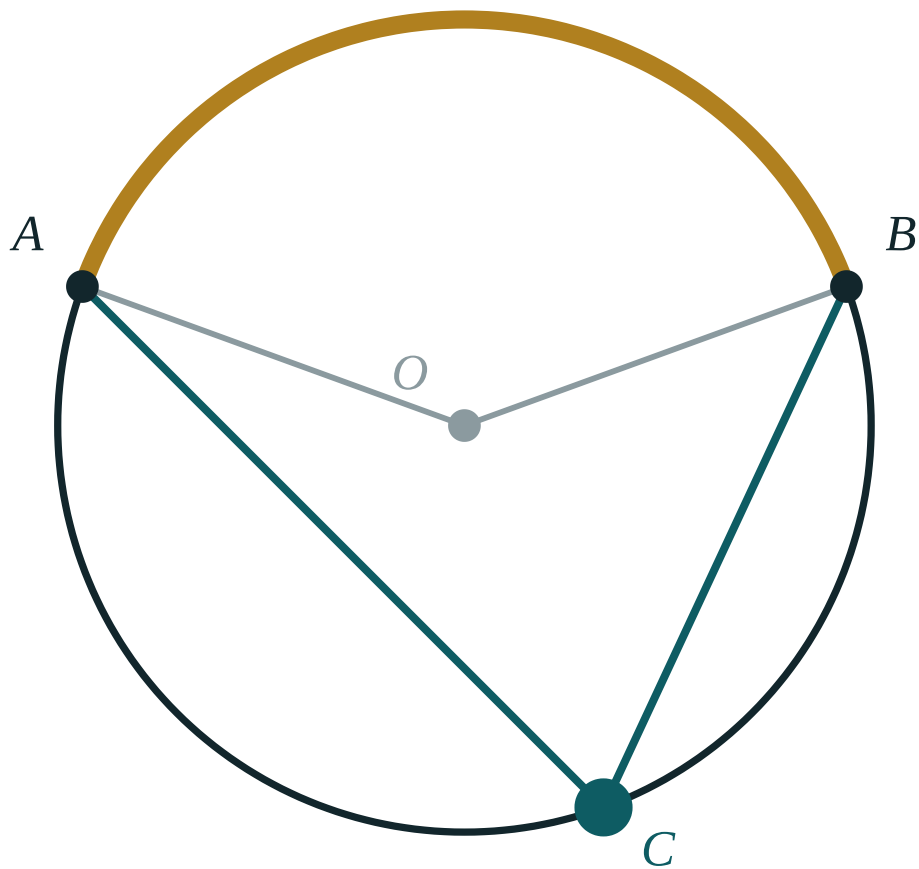
ANSWER

$$5\pi + 20 \approx 35.7$$

03 Interactive model

Cut a length of string equal to the radius and lay it round the rim; the angle it subtends is one radian, and 57° becomes a measurement rather than a fact.

Arc, angle and radius



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Arc	Yoy	Дуга
Arc length	Yoy uzunligi	Длина дуги
Central angle	Markaziy burchak	Центральный угол
Radian	Radian	Радиян
Degree	Gradus	Градус
Sector	Sektor	Сектор
Perimeter of a sector	Sektor perimetri	Периметр сектора
Fraction of a circle	Aylananing ulushi	Доля окружности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

One radian is the angle whose arc equals:

the diameter

the radius

the circumference

π

180° in radians:

$\frac{\pi}{2}$

π

2π

90

l with α in radians:

$R\alpha$

$R^2\alpha$

$\frac{\alpha}{R}$

$2\pi R\alpha$

An arc of 60° in $R = 12$:

The perimeter of a sector is:

135° in radians:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Arc of 90° in $R = 12$

ANSWER 6π

2. Arc of 60° in $R = 12$

ANSWER 4π

3. Arc of 30° in $R = 12$

ANSWER 2π

4. 90° in radians

ANSWER $\frac{\pi}{2}$

5. 60° in radians

ANSWER $\frac{\pi}{3}$

6. π in degrees

ANSWER 180°

7. $R = 5, \alpha = 2: l$

ANSWER 10

Medium · the standard question

7 PROBLEMS

1. 135° in radians

ANSWER $\frac{3\pi}{4}$

2. $\frac{5\pi}{6}$ in degrees

ANSWER 150°

3. $R = 12, \alpha = \frac{\pi}{3} : l$

ANSWER 4π

4. $l = 15, \alpha = 3 : R$

ANSWER 5

5. Sector $R = 10, \theta = 90^\circ$: perimeter

ANSWER $5\pi + 20$

6. Sector $R = 6, \alpha = \frac{\pi}{3}$: perimeter

ANSWER $2\pi + 12$

7. Arc of 120° in $R = 9$

ANSWER 6π

Hard · stretch and reason

7 PROBLEMS

1. An arc of 10 cm subtends 2 rad : the radius

ANSWER 5 cm

2. An arc of 8π in a circle of radius 12 : the angle in degrees

ANSWER 120°

3. A sector of perimeter 20 and radius 6 : its angle in radians

ANSWER $\frac{4}{3}$

4. The minute hand is 12 cm : the arc its tip sweeps in 25 minutes

ANSWER $10\pi\text{ cm}$

5. A pendulum of length 80 cm swings through 12° : the arc

ANSWER $\frac{16\pi}{3}\text{ cm}$

6. Two arcs of equal length in circles of radii 4 and 6 : the ratio of their angles

ANSWER $3 : 2$

7. A wheel of radius 35 cm turns through 4 radians: the distance rolled

ANSWER 140 cm

07 Homework

Homework — 5 tasks

Convert to radians before using $l = R\alpha$, and say so in the working.

- Find the arc length for $\theta = 45^\circ$ in a circle of radius 16 .
- Convert 210° to radians and $\frac{7\pi}{4}$ to degrees.
- A sector has radius 9 and angle $\frac{2\pi}{3}$. Find its arc and perimeter.

4. An arc of 21 cm subtends 1.5 rad . Find the radius.
5. The minute hand of a clock is 10 cm long. Find the arc its tip sweeps in 40 minutes.

The area of a disc

πR^2 — proved by cutting the disc into ever-thinner triangles.

LESSON 46–47

2 × 40 MIN

GEOMETRIYA 9, §28

IGX 7.1

LESSON CLOCK

13'

23'

22'

18'

4'

The proof	13 min
Using the formula	23 min
Rings	22 min
Composite areas	18 min
Homework	4 min

LEARNING OBJECTIVES

- Derive $S = \pi R^2$ from the area of a regular polygon.
- Use $S = \pi R^2$ in both directions.
- Find the area of a ring (annulus).
- Solve problems on composite areas involving circles.

TEXTBOOK

National	pp. 166–171
Cambridge	Core & Extended, pp. 130–136
Workbook	Exercise 7.1

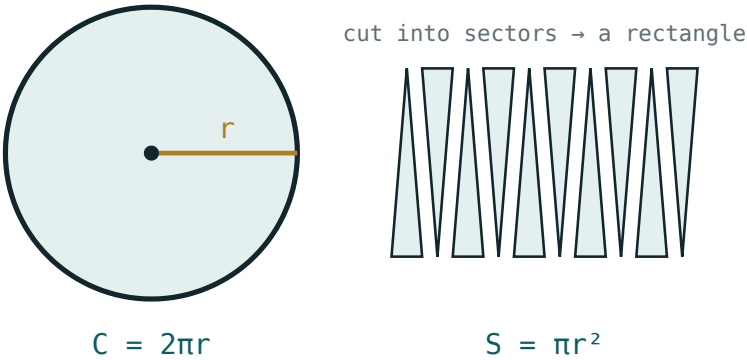
01

Explanation

The proof

A regular n -gon inscribed in the circle has area $S = \frac{1}{2}Pr$. As n grows, P approaches the circumference $2\pi R$ and r approaches R . Hence

$$S \rightarrow \frac{1}{2} \cdot 2\pi R \cdot R = \pi R^2$$



The disc cut into thin sectors and rearranged — the strip is πR long and R high.

The same argument in a picture: cut the disc into many thin sectors and interleave them. The result approaches a rectangle of height R and width πR — half the circumference — whose area is πR^2 .

CIRCUMFERENCE AND AREA ARE ONE FORMULA APART

$C = 2\pi R$ and $S = \pi R^2$, and $\frac{1}{2} CR = S$. Every disc formula in this chapter is a consequence of those two.

Using the formula

$$S = \pi R^2 = \frac{\pi d^2}{4}$$

GIVEN	FIND	WORKING
$R = 7$	S	$49\pi \approx 154$
$d = 10$	S	$25\pi \approx 78.5$
$S = 36\pi$	R	6
$S = 100$	R	$\sqrt{\frac{100}{\pi}} \approx 5.64$

GOING BACKWARDS NEEDS A SQUARE ROOT

$S = 36\pi$ gives $R^2 = 36$ and then $R = 6$ — not $R = 36$. Writing R^2 explicitly before taking the root prevents the slip.

Rings

An **annulus** is the region between two concentric circles of radii R and r :

$$S = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R - r)(R + r)$$

RING	AREA
$R = 10, r = 6$	64π
$R = 5, r = 3$	16π
outer diameter 20, width 2	36π

FACTORISE RATHER THAN SUBTRACT

$\pi(R - r)(R + r)$ is usually quicker than squaring both radii, and it makes “a ring of width w ” questions almost immediate: the area is $\pi w(2r + w)$.

Composite areas

FIGURE	AREA
a semicircle of radius R	$\frac{\pi R^2}{2}$
a quarter-disc of radius R	$\frac{\pi R^2}{4}$
a square of side a minus its inscribed disc	$a^2 - \frac{\pi a^2}{4}$
a disc minus its inscribed square	$\pi R^2 - 2R^2$
a running track: two straights L wide w , two semicircular ends	strips plus rings

The third row is worth remembering as a fraction: an inscribed disc occupies $\frac{\pi}{4} \approx 78.5\%$ of its square, so the corners take about 21.5%.

SUBTRACT, DO NOT GUESS

A shaded region is almost always a difference of two simple areas. Name both, compute both, and subtract — attempting to find the shaded area directly is what makes these questions look hard.

02 Worked examples

EXAMPLE 1 Find the area of a disc of radius 7 cm exactly and to 3 s.f.

① $S = \pi R^2 = 49\pi$
Exact.

② $49 \times 3.14159\dots$

③ ≈ 153.94

④ $\approx 154 \text{ cm}^2$

ANSWER

$49\pi \approx 154 \text{ cm}^2$

EXAMPLE 2 A ring has outer radius 10 and inner radius 6. Find its area.

① $S = \pi(R^2 - r^2)$

② $= \pi(100 - 36)$

③ $= 64\pi$

④ ≈ 201

ANSWER

$64\pi \approx 201$

EXAMPLE 3 A disc is inscribed in a square of side 12. Find the area outside the disc.

- ① Square: 144.
- ② Disc: $R = 6$, $S = 36\pi$.
- ③ $144 - 36\pi$
- ④ $\approx 144 - 113.1 = 30.9$

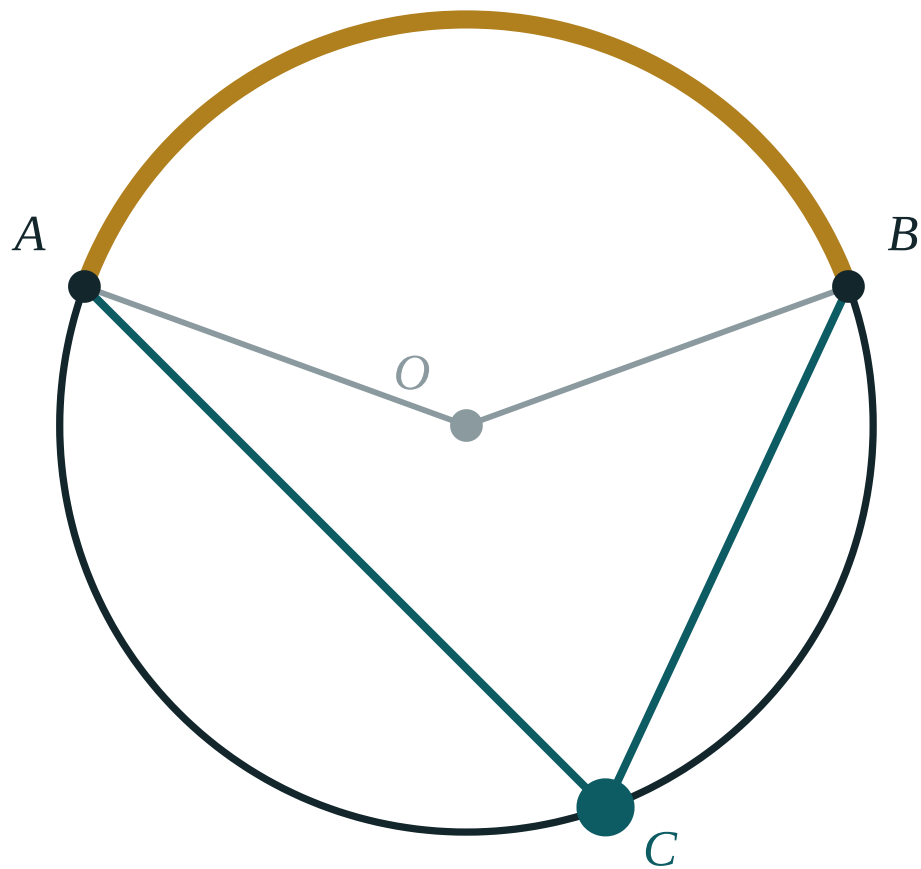
ANSWER

$$144 - 36\pi \approx 30.9$$

03 Interactive model

Cut a paper disc into sixteen sectors and interleave them into a near-rectangle; the class measures its sides as πR and R and the formula is theirs.

From polygon to disc

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Disc	Doira	Круг
Area of a disc	Doira yuzi	Площадь круга
Annulus	Halqa	Кольцо
Ring	Halqa	Кольцо
Composite area	Murakkab yuza	Составная площадь
Shaded region	Bo'yalgan soha	Закрашенная область
Semicircle	Yarim doira	Полукруг
Limit	Limit	Предел

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area of a disc is:

$R = 7$ gives $S =$

$S = 36\pi$ gives $R =$

A ring of radii 10 and 6 :

16π

64π

136π

4π

A disc inscribed in a square occupies:

50%

$\approx 78.5\%$

$\approx 90\%$

100%

A shaded region is usually:

found directly

a difference of two areas

a product

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $R = 7$: S exactly

ANSWER 49π

2. $R = 5$: S exactly

ANSWER 25π

3. $d = 10$: S exactly

ANSWER 25π

4. $S = 36\pi$: R

ANSWER 6

5. $S = 81\pi$: d

ANSWER 18

6. Area of a semicircle, $R = 6$

ANSWER 18π

7. Area of a quarter-disc, $R = 8$

ANSWER 16π

Medium · the standard question

7 PROBLEMS

1. Ring $R = 10, r = 6$

ANSWER 64π

2. Ring $R = 5, r = 3$

ANSWER 16π

3. Square of side 12 minus its inscribed disc

ANSWER $144 - 36\pi$

4. A disc of area 100: its radius

ANSWER ≈ 5.64

5. A ring of outer diameter 20 and width 2

ANSWER 36π

6. A disc minus its inscribed square, $R = 5$

ANSWER $25\pi - 50$

7. Two discs of radii 3 and 4: the radius of a disc of equal total area

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. A disc inscribed in an equilateral triangle of side 12: the area outside it

ANSWER $36\sqrt{3} - 12\pi$

2. A running track: inner radius 30, width 8, straights 100: the tarmac area

ANSWER $800 + 544\pi$

3. A circle whose area equals its circumference numerically: its radius

ANSWER 2

4. A ring of area 96π and inner radius 10: its outer radius

ANSWER 14

5. Three equal circles of radius r touching pairwise: the area between them

ANSWER $r^2(\sqrt{3} - \frac{\pi}{2})$

6. A disc of radius R and one of radius $2R$: the ratio of their areas

ANSWER 1 : 4

7. A goat tethered by 7 m at a corner of a large field: the grazed area

ANSWER $\frac{49\pi}{4}$

07 Homework

Homework — 5 tasks

Give every answer exactly in terms of π and then to three significant figures.

- Find the area of a disc of radius 9 cm.
- A disc has area 64π . Find its radius and circumference.
- Find the area of a ring with radii 12 and 9.
- A disc is inscribed in a square of side 20. Find the area outside the disc.

5. Find the radius of a disc whose area is 200 cm^2 .

The areas of the parts of a disc

Sector and segment — one is a fraction of the disc, the other is that fraction minus a triangle.

LESSON 48–49

2 × 40 MIN

GEOMETRIYA 9, §29

IGX 7.1

LESSON CLOCK

13'

23'

22'

18'

4'

The sector	13 min
The segment	23 min
Shaded regions	22 min
Choosing the form	18 min
Homework	4 min

LEARNING OBJECTIVES

- Find the area of a sector in degrees and in radians.
- Find the area of a segment as sector minus triangle.
- Solve problems on shaded regions built from sectors and segments.
- Choose between the degree and radian forms.

TEXTBOOK

National	pp. 172–178
Cambridge	Core & Extended, pp. 130–140
Workbook	Exercise 7.1

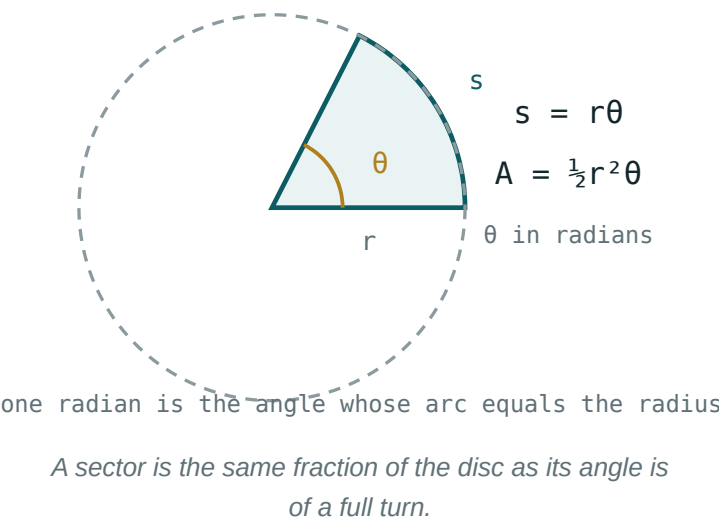
01

Explanation

The sector

A **sector** is the region between two radii and the arc between them — a slice of the disc.

$$S = \frac{\theta}{360^\circ} \cdot \pi R^2 \text{ (degrees)} \quad S = \frac{1}{2} R^2 \alpha \text{ (radians)}$$



θ	FRACTION	AREA, $R = 6$
90°	$\frac{1}{4}$	9π
60°	$\frac{1}{6}$	6π
120°	$\frac{1}{3}$	12π
45°	$\frac{1}{8}$	4.5π

Note the pleasing relation $S = \frac{1}{2} lR$: a sector is “half base times height” with the arc as the base — the same shape of formula as for a triangle.

The segment

A **segment** is the region between a chord and its arc. It is the sector minus the triangle formed by the two radii and the chord:

$$S_{segment} = \frac{1}{2}R^2\alpha - \frac{1}{2}R^2 \sin \alpha = \frac{1}{2}R^2(\alpha - \sin \alpha)$$

with α in radians. In degrees, compute the sector by the fraction and the triangle by $\frac{1}{2}R^2 \sin \theta$, then subtract.

STEP	FORMULA	$R = 10, \Theta = 60^\circ$
sector	$\frac{\theta}{360} \pi R^2$	$\frac{100\pi}{6} \approx 52.36$
triangle	$\frac{1}{2} R^2 \sin \theta$	$50 \sin 60^\circ \approx 43.30$
segment	the difference	≈ 9.06

SIN A NEEDS A IN THE CALCULATOR'S CURRENT MODE

In the radian formula $\frac{1}{2} R^2(\alpha - \sin \alpha)$, both α and $\sin \alpha$ are in radians. Subtracting a degree sine from a radian angle produces nonsense — and it is the standard error in segment questions.

Shaded regions

REGION	BUILT FROM
a segment	sector – triangle
a major sector	disc – minor sector
a lune between two arcs	sector – sector
a square minus four quarter-discs at its corners	$a^2 - \pi r^2$
the region between a square and its inscribed disc	$a^2 - \frac{\pi a^2}{4}$

The fourth row is worth noticing: four quarter-discs of radius r make one whole disc, so no fractions are needed at all.

NAME THE TWO SIMPLE REGIONS BEFORE COMPUTING EITHER

Almost every shaded-area question is “this minus that”, and the difficulty is entirely in seeing which two. Writing the two names on the diagram makes the arithmetic routine.

Choosing the form

IF THE ANGLE IS GIVEN IN	USE
degrees, a nice fraction of 360	$\frac{\theta}{360} \pi R^2$
radians	$\frac{1}{2} R^2 \alpha$
degrees, awkward	convert, then $\frac{1}{2} R^2 \alpha$

The radian form is the one used from Grade 10 onwards, so it is worth becoming fluent in it now even where the degree form would serve.

FOUR FORMULAE, AND THEY ARE ALL THE SAME ONE

$l = \frac{\theta}{360} \cdot 2\pi R = R\alpha$ and $S = \frac{\theta}{360} \cdot \pi R^2 = \frac{1}{2} R^2 \alpha$. Both radian forms come from the degree forms by substituting $\theta = \frac{180\alpha}{\pi}$, and both are shorter.

02 Worked examples

EXAMPLE 1 Find the area of a sector with $R = 6$ and $\theta = 120^\circ$.

$$1 \quad \frac{120}{360} = \frac{1}{3}$$

$$2 \quad \pi R^2 = 36\pi$$

$$3 \quad S = \frac{1}{3} \times 36\pi$$

$$4 \quad = 12\pi \approx 37.7$$

ANSWER

$$12\pi \approx 37.7$$

EXAMPLE 2 Find the area of the segment cut off by a chord subtending 60° in a circle of radius 10.

① Sector: $\frac{1}{6} \times 100\pi \approx 52.36$.

② Triangle: $\frac{1}{2} \times 100 \times \sin 60^\circ \approx 43.30$.
Equilateral, in fact.

③ $52.36 - 43.30$

④ ≈ 9.06

ANSWER

≈ 9.06

EXAMPLE 3 A sector has area 24π and radius 12. Find its angle in degrees.

① $\frac{\theta}{360} \times 144\pi = 24\pi$

② $\frac{\theta}{360} = \frac{24}{144} = \frac{1}{6}$

③ $\theta = 60^\circ$

④ In radians, $\frac{\pi}{3}$.

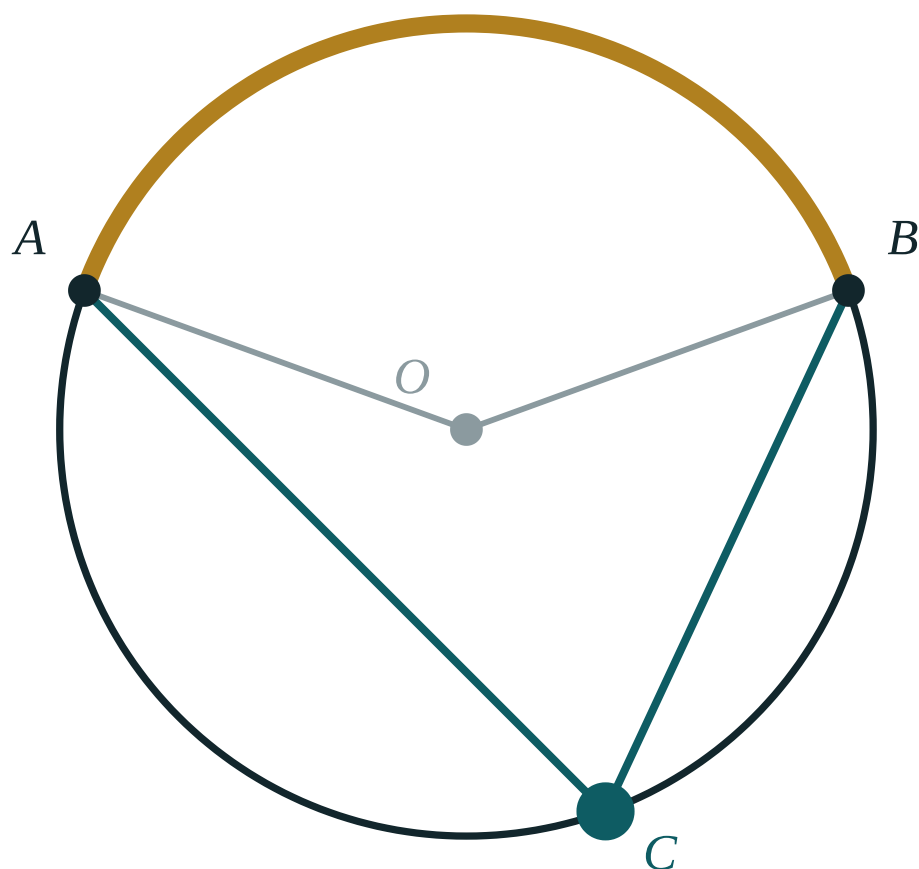
ANSWER

60°

03 Interactive model

Draw one circle with a 60° chord and shade the segment; ask which two regions the class would subtract before writing a single number.

Sector and segment



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Minor sector	Kichik sektor	Малый сектор
Major sector	Katta sektor	Большой сектор
Chord	Vatar	Хорда
Shaded region	Bo'yalgan soha	Закрашенная область
Subtract	Ayirmoq	Вычесть
Exact answer	Aniq javob	Точный ответ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A sector's area in degrees:

$$\frac{\theta}{360}\pi R^2$$

$$\frac{\theta}{180}\pi R^2$$

$$\theta\pi R^2$$

$$\frac{\theta}{360}2\pi R$$

A sector's area in radians:

$$R^2\alpha$$

$$\frac{1}{2}R^2\alpha$$

$$2R^2\alpha$$

$$\frac{1}{2}R\alpha$$

A segment is:

sector + triangle

sector – triangle

disc – sector

half a sector

$R = 6$, $\theta = 120^\circ$: the sector area

A sector of area 24π and $R = 12$ has angle:

$S = \frac{1}{2}lR$ resembles the formula for:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Sector $R = 6, \theta = 90^\circ$

ANSWER 9π

2. Sector $R = 6, \theta = 60^\circ$

ANSWER 6π

3. Sector $R = 6, \theta = 120^\circ$

ANSWER 12π

4. Sector $R = 4, \alpha = \frac{\pi}{2}$

ANSWER 4π

5. Sector $R = 10, \alpha = 1$

ANSWER 50

6. Area of a semicircle, $R = 8$

ANSWER 32π

7. A segment is sector minus

ANSWER A triangle

Medium · the standard question

7 PROBLEMS

1. Segment, $R = 10$, $\theta = 60^\circ$

ANSWER ≈ 9.06

2. Segment, $R = 6$, $\theta = 90^\circ$

ANSWER $9\pi - 18 \approx 10.3$

3. Sector of area 24π , $R = 12$: the angle

ANSWER 60°

4. Sector of area 50 , $R = 10$: α

ANSWER 1 rad

5. Major sector, $R = 6$, $\theta = 120^\circ$

ANSWER 24π

6. A square of side 10 minus four quarter-discs of radius 5

ANSWER $100 - 25\pi$

7. Segment, $R = 8$, $\alpha = \frac{\pi}{3}$

ANSWER $\frac{32\pi}{3} - 16\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. A segment of $R = 12$ and $\theta = 90^\circ$

ANSWER $36\pi - 72 \approx 41.1$

2. A chord of length R : the area of the minor segment

ANSWER $R^2(\frac{\pi}{6} - \frac{\sqrt{3}}{4})$

3. A goat tethered by 12 m at the corner of a square shed, grazing a 90° region

ANSWER 36π

4. Two circles of radius r whose centres are r apart: the area of overlap

ANSWER $r^2(\frac{2\pi}{3} - \frac{\sqrt{3}}{2})$

5. A sector of perimeter 28 and radius 8 : its area

ANSWER 48

6. The greatest sector area for a given perimeter P

ANSWER $\frac{P^2}{16}$, at $\alpha = 2$

7. A disc of radius 10 with a 120° sector removed: the remaining area

ANSWER $\frac{200\pi}{3}$

07 Homework

Homework — 5 tasks

For a segment, write down the sector and the triangle separately before subtracting.

- Find the area of a sector with $R = 9$ and $\theta = 40^\circ$.
- Find the area of a sector with $R = 5$ and $\alpha = 1.2$.
- Find the area of the segment cut off by a 90° chord in a circle of radius 8 .

4. A sector has area 27π and radius 9. Find its angle.
5. A square of side 12 has a quarter-disc of radius 12 drawn from one corner. Find the area outside the quarter-disc.

Chapter exercises — Cambridge solids, surface area and volume

The circle formulae carried into three dimensions, as an IGCSE paper uses them.

LESSON 50

1 × 40 MIN

GEOMETRIYA 9, III BOB MASHQLARI

IGX 7.2–7.3

LESSON CLOCK

8'

14'

12'

6'

The formulae	8 min
Where the circle appears	14 min
Rearranging	12 min
Homework	6 min

LEARNING OBJECTIVES

- Use the volume and surface-area formulae for a prism, cylinder, cone and sphere.
- Recognise which circle formula each one contains.
- Solve problems requiring a formula to be rearranged.
- Give answers in terms of π and to three significant figures.

TEXTBOOK

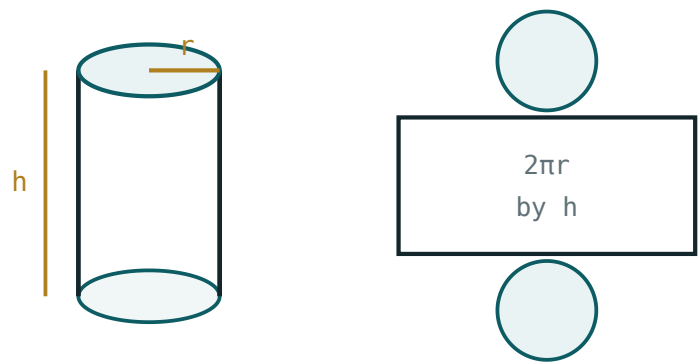
National	pp. 179–182
Cambridge	Core & Extended, pp. 137–152
Workbook	Exercise 7.2–7.3

01

Explanation

The formulae

SOLID	VOLUME	CURVED SURFACE	TOTAL SURFACE
prism	$A \times h$	perimeter $\times h$	+ 2A
cylinder	$\pi R^2 h$	$2\pi R h$	$2\pi R h + 2\pi R^2$
cone	$\frac{1}{3}\pi R^2 h$	$\pi R L$	$\pi R L + \pi R^2$
sphere	$\frac{4}{3}\pi R^3$	$4\pi R^2$	$4\pi R^2$
pyramid	$\frac{1}{3}Ah$	—	—



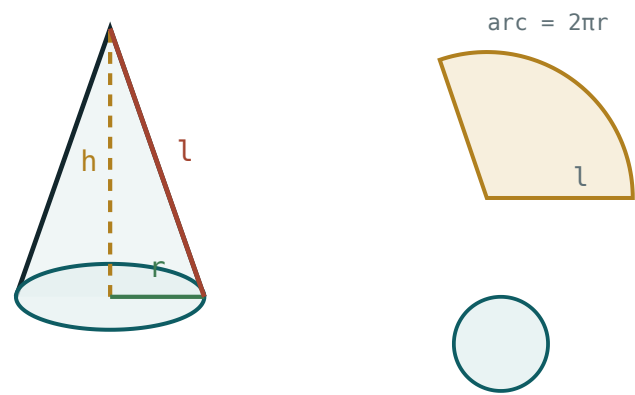
$S = 2\pi r^2 + 2\pi rh$ – two circles and one rectangle
The net of a cylinder — a rectangle $2\pi R$ wide and two discs.

THE CONE AND THE PYRAMID SHARE THE $\frac{1}{3}$

Anything that tapers to a point has one third the volume of the prism or cylinder on the same base with the same height. That single sentence covers both rows.

Where the circle appears

FORMULA	CONTAINS	WHY
$V = \pi R^2 h$	πR^2	the area of the circular base
$2\pi R h$	$2\pi R$	the circumference, unrolled
$\pi R L$	πR	half the circumference times L
$L^2 = R^2 + h^2$	Pythagoras	the slant height of a cone



$S = \pi r^2 + \pi r l$ – a circle and a sector

The net of a cone — a sector whose arc is the base circumference.

The curved surface of a cone unrolls into a sector of radius L whose arc is $2\pi R$; its area is $\frac{1}{2} \times 2\pi R \times L = \pi RL$, by the sector formula of the last lesson.

Rearranging

GIVEN	FIND	WORKING
a cylinder, $V = 500, R = 5$	h	$\frac{500}{25\pi} \approx 6.37$
a sphere, $V = 36\pi$	R	3
a cone, $R = 6, h = 8$	L , then the surface	$L = 10, S = 96\pi$
a sphere, $S = 100\pi$	R	5

A CUBE ROOT FOR A SPHERE’S VOLUME

$V = \frac{4}{3}\pi R^3 = 36\pi$ gives $R^3 = 27$, so $R = 3$. Taking a square root instead is the usual slip.

02 Worked examples

EXAMPLE 1 A cylinder has radius 5 and height 12. Find its volume and total surface area.

$$① \quad V = \pi R^2 h = \pi \times 25 \times 12$$

$$② \quad = 300\pi \approx 942$$

$$③ \quad S = 2\pi Rh + 2\pi R^2 = 120\pi + 50\pi$$

$$④ \quad = 170\pi \approx 534$$

ANSWER

$$V = 300\pi, S = 170\pi$$

EXAMPLE 2 A cone has radius 6 and height 8. Find its slant height, volume and total surface area.

$$① \quad L = \sqrt{36 + 64} = 10$$

Pythagoras.

$$② \quad V = \frac{1}{3}\pi \times 36 \times 8 = 96\pi$$

$$③ \quad S = \pi RL + \pi R^2 = 60\pi + 36\pi$$

$$④ \quad = 96\pi \approx 302$$

ANSWER

$$L = 10, V = 96\pi, S = 96\pi$$

EXAMPLE 3 A sphere has volume 36π . Find its radius and surface area.

$$\textcircled{1} \quad \frac{4}{3}\pi R^3 = 36\pi$$

$$\textcircled{2} \quad R^3 = 27$$

$$\textcircled{3} \quad R = 3$$

A cube root.

$$\textcircled{4} \quad S = 4\pi R^2 = 36\pi \approx 113$$

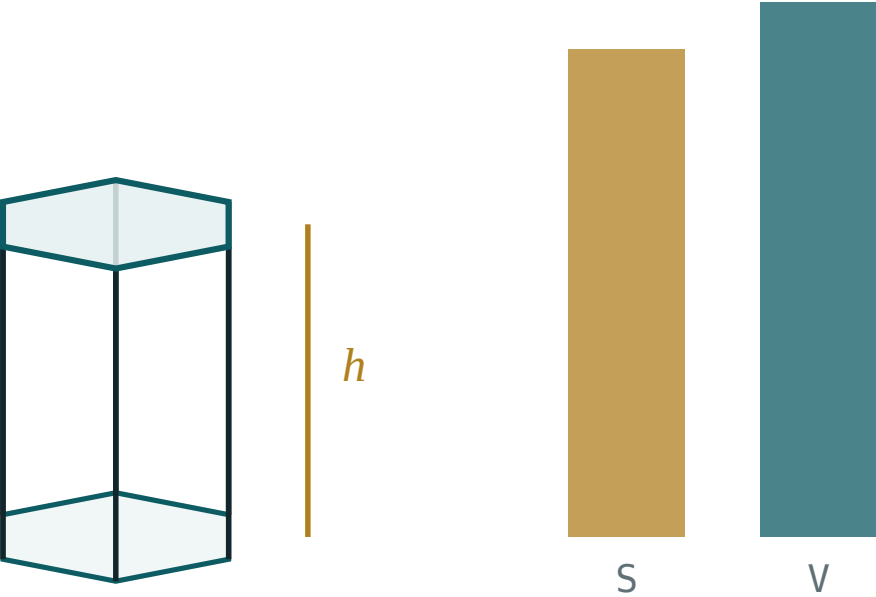
ANSWER

$$R = 3, S = 36\pi$$

03 Interactive model

Fill a cone with sand and pour it into a cylinder of the same base and height; three pours fill it, and the one third is measured rather than asserted.

Volumes of the standard solids



base area B **41.6**

perimeter P **24**

total S **227.1**

volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Sphere	Sfera	Сфера
Slant height	Yasovchi	Образующая
Curved surface area	Yon sirt yuzi	Площадь боковой поверхности
Cross-section	Kesim	Сечение
Net	Yoyilma	Развёртка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a cylinder:

$$2\pi Rh$$

$$\pi R^2 h$$

$$\frac{1}{3} \pi R^2 h$$

$$\frac{4}{3} \pi R^3$$

The curved surface of a cylinder:

$$\pi R^2$$

$$2\pi Rh$$

$$\pi RL$$

$$4\pi R^2$$

The volume of a cone:

$$\pi R^2 h$$

$$\frac{1}{3} \pi R^2 h$$

$$\frac{1}{2} \pi R^2 h$$

$$\pi RL$$

The slant height of a cone:

$$R + h$$

$$\sqrt{R^2 + h^2}$$

$$\sqrt{R^2 - h^2}$$

$$Rh$$

The surface area of a sphere:

$$\pi R^2$$

$$2\pi R^2$$

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$V = 36\pi$ for a sphere gives $R =$

$$3$$

$$6$$

$$9$$

$$27$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Cylinder $R = 5, h = 12$: V

ANSWER 300π

2. Cylinder $R = 5, h = 12$: curved surface

ANSWER 120π

3. Cone $R = 6, h = 8$: L

ANSWER 10

4. Cone $R = 6, h = 8$: V

ANSWER 96π

5. Sphere $R = 3$: V

ANSWER 36π

6. Sphere $R = 3$: S

ANSWER 36π

7. Volume of a prism

ANSWER $A \times h$

Medium · the standard question

7 PROBLEMS

1. Cylinder $R = 5$, $h = 12$: total surface

ANSWER 170π

2. Cone $R = 6$, $h = 8$: total surface

ANSWER 96π

3. Sphere of volume 36π : R

ANSWER 3

4. Sphere of surface 100π : R

ANSWER 5

5. Cylinder of volume 500 and $R = 5$: h

ANSWER ≈ 6.37

6. Cone of volume 100π and $R = 5$: h

ANSWER 12

7. A hemisphere of radius 6: its volume

ANSWER 144π

Hard · stretch and reason

7 PROBLEMS

1. A sphere fits exactly in a cylinder: the ratio of their volumes

ANSWER $2 : 3$

2. And of their curved surfaces

ANSWER $1 : 1$

3. A cone, a hemisphere and a cylinder of the same radius and height R

ANSWER $1 : 2 : 3$

4. A cylinder of height h and volume V : its radius

ANSWER $\sqrt{\frac{V}{\pi h}}$

5. A cone of slant height 13 and radius 5: its volume

ANSWER 100π

6. A sphere of radius R melted into cones of radius R and height R

ANSWER 4 cones

7. A cylindrical tank of diameter 2 m holding 6280 L: its height

ANSWER ≈ 2 m

07 Homework

Homework — 5 tasks

Give every answer in terms of π and then to three significant figures, with units.

1. A cylinder has radius 7 and height 10. Find its volume and total surface area.
2. A cone has radius 9 and height 12. Find its slant height and total surface area.
3. A sphere has surface area 144π . Find its radius and volume.
4. A cone of volume 48π has radius 6. Find its height.

5. Show that a sphere inscribed in a cylinder occupies two thirds of its volume.

Control work 3, and work on the mistakes

Regular polygons and the circle in one paper, and the chapter closed.

LESSON 51–52

2 × 40 MIN

GEOMETRIYA 9, NAZORAT ISHI 3

IGX 3, 7 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Use the interior and exterior angle formulae under time.
- Relate R , r and a for a regular polygon.
- Compute arc, sector and segment accurately.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 121–182
Cambridge	Core & Extended, pp. 58–152
Workbook	Control paper G3

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	A regular polygon has an interior angle of 150° : find n	4	L37
2	A cyclic quadrilateral has $\angle A = 78^\circ$: find $\angle C$	3	L33–34
3	A regular hexagon of side 8: find R , r and its area	6	L38–41
4	A circle of radius 15: find the arc and sector for $\theta = 72^\circ$	6	L44–47
5	A ring with radii 13 and 5: find its area	5	L46–47
6	The segment cut off by a 90° chord in a circle of radius 10	6	L48–49

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for going through the exterior angle; Q3 two for $R = a$; Q4 one for the fraction $\frac{72}{360}$; Q5 two for factorising; Q6 two for subtracting the triangle.

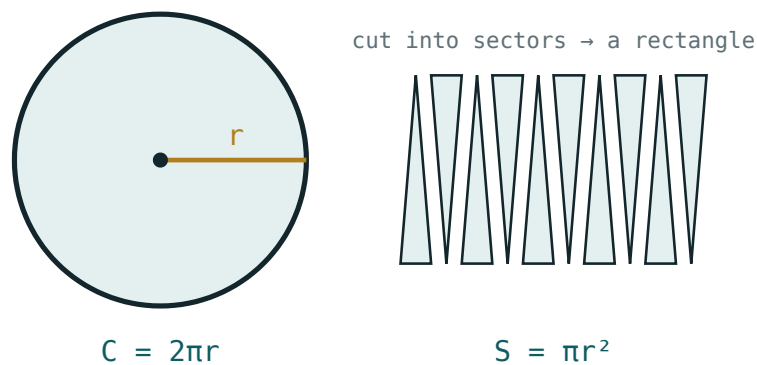
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
interior-angle equation solved directly	$\frac{(n - 2)180}{n} = 150$	$e = 30 \Rightarrow n = 12$
cyclic angles taken as equal	$\angle C = 78^\circ$	$\angle C = 102^\circ$
R	$R = \frac{a}{2}$	$R = a \text{ for a hexagon}$
circumference used for area	$2\pi R$	πR^2
ring radii squared wrongly	$\pi(13 - 5)^2$	$\pi(13^2 - 5^2)$
segment left as the sector	25π	$25\pi - 50$
degrees used in $\frac{1}{2}R^2\alpha$	$\frac{1}{2} \times 100 \times 90$	convert to radians first

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter III as one map

BLOCK	THE SENTENCE
inscribed polygons	vertices on the circle; a quadrilateral needs $\angle A + \angle C = 180^\circ$
circumscribed polygons	sides touching the circle; a quadrilateral needs $a + c = b + d$
regular polygons	$e = \frac{360^\circ}{n}, i = 180^\circ - e$
the two radii	$a = 2R \sin \frac{180^\circ}{n}, r = R \cos \frac{180^\circ}{n}$
the circumference	$C = 2\pi R$
the arc	$l = \frac{\theta}{360} \cdot 2\pi R = R\alpha$
the disc	$S = \pi R^2$
the sector	$S = \frac{\theta}{360} \cdot \pi R^2 = \frac{1}{2} R^2 \alpha$
the segment	sector – triangle



One circle carries every formula of the chapter.

LOOKING FORWARD

Chapter IV returns to similarity, in its most computational form: proportional segments in a triangle and in a circle, the mean proportional, and the constructions that produce it. The chapter after this one is Grade 10.

02 Worked examples

EXAMPLE 1 Model answer, Q3: a regular hexagon of side 8.

① Central angle 60° — the triangles are equilateral.

② $R = a = 8$
The key fact.

③ $r = R \cos 30^\circ = 4\sqrt{3}$

④ $S = \frac{1}{2} \times 48 \times 4\sqrt{3} = 96\sqrt{3} \approx 166$

ANSWER

$$R = 8, r = 4\sqrt{3}, S = 96\sqrt{3}$$

EXAMPLE 2 Model answer, Q4: $R = 15$, $\theta = 72^\circ$.

① $\frac{72}{360} = \frac{1}{5}$
The fraction.

② $l = \frac{1}{5} \times 30\pi = 6\pi \approx 18.8$

③ $S = \frac{1}{5} \times 225\pi = 45\pi$

④ ≈ 141

ANSWER

$$l = 6\pi, S = 45\pi$$

EXAMPLE 3 Model answer, Q6: the segment for $\theta = 90^\circ$, $R = 10$.

① Sector: $\frac{1}{4} \times 100\pi = 25\pi \approx 78.5$.

② Triangle: $\frac{1}{2} \times 10 \times 10 = 50$.
Right-angled.

③ $25\pi - 50$

④ ≈ 28.5

ANSWER

$$25\pi - 50 \approx 28.5$$

03 Interactive model

Put a Q6 answer of 25π on the board and ask what is still missing; the class names the triangle before you do.

Chapter III in twelve questions

An inscribed polygon has:

sides touching

vertices on the circle

the circle inside

no circle

A cyclic quadrilateral satisfies:

$$\angle A = \angle C$$

$$\angle A + \angle C = 180^\circ$$

$$a + c = b + d$$

nothing

A tangential quadrilateral satisfies:

$$\angle A + \angle C = 180^\circ$$

$$a + c = b + d$$

$$a = c$$

nothing

The incentre is where the _____ meet:

medians

angle bisectors

perpendicular bisectors

altitudes

An exterior angle of a regular n -gon:

$$\frac{180^\circ}{n}$$

$$\frac{360^\circ}{n}$$

$$\frac{(n-2)180^\circ}{n}$$

$$n \cdot 180^\circ$$

For a regular hexagon:

$$R = 2a$$

$$R = a$$

$$R = \frac{a}{2}$$

$$R = a\sqrt{2}$$

a equals:

$$2R \sin \frac{180^\circ}{n}$$

$$2R \cos \frac{180^\circ}{n}$$

$$R \sin \frac{360^\circ}{n}$$

$$R \tan \frac{180^\circ}{n}$$

C equals:

$$\pi R$$

$$2\pi R$$

$$\pi R^2$$

$$2\pi R^2$$

An arc with α in radians:

$$R\alpha$$

$$R^2\alpha$$

$$2\pi R\alpha$$

$$\frac{\alpha}{R}$$

The area of a disc:

$$2\pi R$$

$$\pi R^2$$

$$\pi d^2$$

$$\frac{4}{3}\pi R^3$$

A sector in radians:

$$R^2\alpha$$

$$\frac{1}{2}R^2\alpha$$

$$2R^2\alpha$$

$$\frac{1}{2}R\alpha$$

A segment is:

sector + triangle

sector – triangle

disc – sector

half a disc

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Regular polygon	Muntazam ko'pburchak	Правильный многоугольник
Inscribed circle	Ichki chizilgan aylana	Вписанная окружность
Circumscribed circle	Tashqi chizilgan aylana	Описанная окружность
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is quickest through:

the interior angle

the exterior angle

the angle sum

a diagram

In Q2 the angles are:

equal

supplementary

complementary

unrelated

In Q3, R equals:

$$\frac{a}{2}$$

$$a$$

$$a\sqrt{2}$$

$$2a$$

Q4 begins with:

the circumference

the fraction

the radius

a sketch

Q5 is quickest by:

squaring separately

factorising

a calculator

estimating

Q6 must subtract:

a sector

a triangle

a ring

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Interior angle 150° : n

ANSWER 12

2. Cyclic, $\angle A = 78^\circ$: $\angle C$

ANSWER 102°

3. Hexagon of side 8: R

ANSWER 8

4. Hexagon of side 8: r

ANSWER $4\sqrt{3}$

5. $R = 15$, $\theta = 72^\circ$: the arc

ANSWER 6π

6. $R = 15$, $\theta = 72^\circ$: the sector

ANSWER 45π

7. Ring 13 and 5: its area

ANSWER 144π

Medium · the standard question

7 PROBLEMS

1. Hexagon of side 8: its area

ANSWER $96\sqrt{3}$

2. Segment, $R = 10$, $\theta = 90^\circ$

ANSWER $25\pi - 50$

3. Exterior angle 30° : n

ANSWER 12

4. A square of side 10: R

ANSWER $5\sqrt{2}$

5. Circumference of a circle of radius 15

ANSWER 30π

6. Area of that circle

ANSWER 225π

7. A sector of $R = 15$ and $\alpha = 1.2$

ANSWER 135

Hard · stretch and reason

7 PROBLEMS

1. A regular polygon with $R = 10$ and $a = 10$: its area

ANSWER $150\sqrt{3}$

2. A ring of area 144π and inner radius 5: its outer radius

ANSWER 13

3. A segment of $R = 6$ and $\theta = 120^\circ$

ANSWER $12\pi - 9\sqrt{3}$

4. A cyclic quadrilateral with $\angle A : \angle C = 5 : 7$

ANSWER 75° and 105°

5. A regular octagon of side 6: its area

ANSWER ≈ 174

6. A circle inscribed in a regular hexagon of side 8: its area

ANSWER 48π

7. The area between that circle and the hexagon

ANSWER $96\sqrt{3} - 48\pi$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before Chapter IV begins.

1. A regular polygon has an exterior angle of 15° . Find n and its interior angle.
2. A regular hexagon has side 10. Find R , r and its area.
3. A circle has radius 12. Find the arc and the sector area for $\theta = 150^\circ$.
4. Find the area of a ring with radii 17 and 8.

5. Find the area of the segment cut off by a 60° chord in a circle of radius 12.